TEAM CONTRACT

GNG1103, Section # B00 Team # 5 Team Name Equilibrium

Team Members:

1) Cameron Kibble

2) Alex Mou

3) Shyaan Qureshi

4) Juliana Sangregorio

5) Christopher Ly

Team Procedures

1. Day, time, and place for regular **team meetings**:

Thursdays, 1:15pm
(STM) Lobby outside of the Makerspace
150 Louis-Pasteur Private #107, Ottawa, ON K1N 6N5

2. Preferred method of **communication** (e.g. e-mail, cell phone, Facebook, Brightspace Discussion Board, face-to-face, in a certain class) in order to discuss the project and to inform each other of team meetings, announcement, updates, reminders, problems:

We will use our phones in a messages group chat for communication. All important information will be posted on Trello.

3. **Decision-making policy** (by consensus? by majority vote?):

To further inspire and foster collaborative mindsets, we will be using a consensus system, everyone's input will be weighed equally. The team will then move forward with refining the options and decisions that need to be made.

4. Method for setting and following meeting **agendas** (Who will set each agenda? When? How will team members be notified/reminded? Who will be responsible for the team following the agenda during a team meeting? What will be done to keep the team on track during a meeting?):

The project manager of the given week will set the meeting agenda 24 hours prior. The team members will be notified using messages. In order to keep the team on track, we will have a list of tasks that should be completed during the meeting. When the meeting concludes, the role rotation matrix will begin its next period.

5. Method of **record keeping** (Who will be responsible for recording & disseminating minutes? How & when will the minutes be disseminated? Where will all agendas & minutes be kept?):

   Meeting minutes will be recorded and summarized via the GoodNotes program and sent out after the meeting, as well as linked in the Trello board week by week.
   Link:
   https://trello.com/invite/b/6aa9ad2a3a3e7f9beb60d640/ATTIbdec62b60d596b8ccabdf1274b3aef224B1DDA9F/gng1103-project

6. Member role and responsibilities, and role rotation matrix in the project team: To foster the development of a diverse skill set among your project team members, it is expected that each member contributes to all facets of the project, rather than specializing in a single area. One effective approach to achieving this objective is to implement a weekly role rotation system within the team, providing each member with the opportunity to experience and perform the various roles and responsibilities. To facilitate this process, please refer to the "**Role Rotation Matrix of Members**" document and generate a role rotation matrix for your project team members below.

Member Role Rotation Matrix in a Project Team of 5 members													
			Semester Week										
Name	**Role**	**Responsibilities**	**W1**	**W2**	**W3**	**W4**	**W5**	**W6**	**W7**	**W8**	**W9**	**W10**	**W11**	**W12**
Christopher	Designer (D)	Define visual aspects, interface, appearance, and usability criteria of the product.	D	D	P	P	Q	Q	E	E	R	R	D	D
Juliana	Engineer (E)	Define product function, subfunctions, manufacturing technique, carry out feasibility study, analyze prototype, and	E	E	R	R	D	D	P	P	Q	Q	E	E

		test prototypes.												
Shyaan	Project Manager (P)	Plan, organize, communicate, and track project tasks, deadlines, and milestones.	P	P	Q	Q	E	E	R	R	D	D	P	P
Alex	Report writer & Marketer (R)	Write project deliverables, identify clients/users, promote, and sell ideas and product.	R	R	D	D	P	P	Q	Q	E	E	R	R
Cameron	Quality Controller (Q)	Proofread project deliverables, identify/define quality standards, check prototype and test quality.	Q	Q	E	E	R	R	D	D	P	P	Q	Q

7. Working in a team can sometimes lead to **conflicts**. If conflicts arise in your team, how do you plan to address and resolve them? What is your conflict resolution strategy?

There are many different styles of conflict management, especially depending on the type of conflict at hand. Empathy is a core part of all conflict management, as one must be able to understand how others are feeling in order to address the root issue. Active listening and empathizing with the problem statement from all parties often de-escalate the situation on its own. Should these strategies not work, we will involve a TA to assist in being a mediator.

Team Expectations

Work Quality

1. **Project standards** (What is a realistic level of quality for team presentations, collaborative writing, individual research, preparation of drafts, peer reviews, etc.?):

We will complete the assigned work minimum two days before the final deadline so that the team has one full day to review, provide feedback, and make necessary revisions. Team presentations and collaborative writing should be clear, organized, accurate, and consistent with the project requirements. Individual research should use reliable sources and be properly cited, with information checked for accuracy before being included in the final work. Drafts should be

complete enough for meaningful peer review, and each member is expected to review other members' work and provide constructive feedback before final submission.

2. **Strategies** to fulfill these standards:

We will break larger assignments into smaller tasks and assign each task to a specific member through the Trello board. Each member will complete their assigned work at least two days before the final deadline whenever possible. This allows the team to review, proofread, and make necessary changes at least one day before submission. Research will be supported by reliable and properly cited sources, and members will communicate any uncertainty or missing information to the group rather than leaving incomplete work for the rest of the team. The Quality Controller will review the final work for consistency, formatting, accuracy, and completeness before submission.

Team Participation

1. Strategies to ensure cooperation and equal distribution of tasks:

Tasks will be divided based on workload and tracked through Trello. Each member will have clear responsibilities and deadlines. The weekly role rotation will ensure responsibilities are shared throughout the project. Tasks will be redistributed if one member has significantly more work.

2. Strategies for encouraging/including ideas from all team members (team maintenance):

We will have an opportunity to contribute during meetings. We will listen to and consider everyone's ideas before reaching a consensus. Those who have not contributed will be given an opportunity to share their input.

3. Strategies for keeping on task (task maintenance):

The Project Manager will keep meetings focused on the agenda, deadlines, and outstanding tasks. Trello will track assignments and progress. Before each meeting ends, the team will confirm what needs to be completed, who is responsible, and when it is due.

4. Preferences for leadership (informal, formal, individual, shared):

Leadership will be shared through the weekly role rotation. The Project Manager will organize meetings and track progress, while major decisions will be made through consensus. All members remain responsible for the project.

Personal Accountability

1. Expected individual attendance, punctuality, and participation at all team meetings:

We are expected to attend Thursday meetings at 1:15pm in the STM lobby outside the Makerspace and arrive prepared to participate. If unable to attend, members must notify the group chat as early as possible and remain responsible for their assigned work.

2. Expected level of responsibility for fulfilling team assignments, timelines, and deadlines:

Members are responsible for completing their assigned work by the agreed deadline and keeping Trello updated. If a deadline cannot be met, the member must notify the team 24 hours before the deadline.

3. Expected level of communication with other team members:

Members are expected to regularly check the group chat and communicate project updates, problems, availability, and requests for help. Important decisions should be communicated to the entire team and posted on trello.

4. Expected level of commitment to team decisions and tasks:

Once a decision is reached through consensus, members are expected to support it and complete their assigned responsibilities. If new information requires a change, the decision will be discussed with the entire team.

Consequences for Failing to Follow Procedures and Fulfill Expectations

1. Describe, as a group, how you would handle **infractions** of any of the obligations of this team contract:

   We would start by having a group meeting to discuss what went wrong; we would remind each other of our responsibilities and work as a team to find solutions. We could always help each other if someone is having a harder time with some of the tasks in order to make sure that we are staying on track.

2. Describe what your team will do **if the infractions continue**:

   We would hold another meeting and once again try to discuss possible solutions and remind each other of consequences when there are infractions. If we still can't reach a solution, we would consult with the TA or PM and ask for their input.

****

a) *I participated in formulating the standards, roles, and procedures as stated in this contract.*
b) *I understand that I am obligated to abide by these terms and conditions.*
c) *I understand that if I do not abide by these terms and conditions, I will suffer the consequences as stated in this contract.*

1) Alex Mou date: 9/26/2026

2) Cameron Kibble date: 9/27/2026

3) Christopher Ly date: 9/27/2026

4) Juliana Sangregorio date: 9/27/2026

5) Shyaan Qureshi date: 9/27/2026

01:06:31
report generated or dashboard for high level
01:06:37
inside that that we can see so like volume tracking tracking appointments, you know volume by the provider
01:06:45
surface and location friction matrix monitor the cancellation rate or no show.
01:06:51
and average wedding time at the clinics.
01:06:54
utilization, better weightless, fulfillment speed and overall
01:07:00
facility utilization
01:07:06
and
01:07:14
needs a lot of compliance with India and India and there are many other
01:07:21
and the healthcare domain.
01:07:26
In terms of the data collection, we want minimum collection, correct minimum necessary info and avoid the tail clinical history.
01:07:36
and encryption and audits, secure authentication,
01:07:42
in transit and audit logs.
01:07:47
and comply with PIPIDA and Ontario
01:07:53
I think there may be more compliance and legal readiness that we will check on the way but this is what I get
01:08:08
go back
01:08:11
the clinical system, electronic medical record replacement, we are not doing that clinical notes.
01:08:18
take prescription or lactoseps, that's not in this goal.
01:08:22
Yeah, administrative express, low health card verification, we are trying to verify the health card or direct billing or insurance claim or tally health videos, so that's not in this code.
01:08:38
clinical through MBB success criteria, post title.
01:08:44

clean setup completed in under an hour without manual spreadsheet so we should be able to migrate their access data out of the
01:08:55
on the paper like how we gonna on
01:08:59
in an arm or so we have to have that strategy probably we have to understand.
01:09:07
you know, efficient, you know, many we can do that, yeah, there are multiple ways to do that.
01:09:16
efficient sense managed booking via Android and iOS
01:09:21
successfully because we are
01:09:26
Reliable flow, real time check is weightless and automated reminder
01:09:32
functional, reliable.
01:09:39
from cleaning operation to mobile quality.
01:09:43
So this is all you know till here.
01:09:49
and the student design challenges on
01:09:54
board and senior city house simple and initial
01:10:02
minimum patient data is required for
01:10:06
how can you know the
01:10:11
be prevented across the globe.
01:10:14
That's the manual code.
01:10:17
So now that was you know all about the Cleaning Flow Project and now I'm going to talk a bit about the
01:10:29
basically this is a chrome extension we have already built it you know it's already in the Google
01:10:38
you can download and check but there are still a lot of bugs and I think
01:10:48
move on it.
01:10:51
two projects are connected.

01:10:53
We are going to build up the cleaning floor, right? We are going to build up that platform that we need to check the responsive
01:10:59
design that it works at least on desktop or iPad or Samsung tabs. So we have to validate that on the real devices.
01:11:09
Also the mobile apps that we are building with clinical tech, those apps need to be tested on the real device for Android and iOS.
01:11:20
From the domain point of view, we have complete linear-friendly flow. It's like, you know, it's into the healthier domain and project or product that we want to build. And mobile keyway checker is completely untapped, but that is what, you know, that is what
01:11:36
We do at Kayak building the tech tools at the company and other projects that are new initiatives that we want to launch as a new company and help potential
01:11:49
future entrepreneurs who want to start their career as an entrepreneur and we use these projects to help them to start their career. So it could be you as well, so it's not about just developing a program by
01:12:04
in the company, you know, with that product and
01:12:07
leading our business, going beyond the development disco.
01:12:12
about like marketing sales, operation and daily people or the clinics pay for it.
01:12:19
Okay, so I will have a question with
01:12:23
I could now and I would
01:12:26
you answer any questions that you may have after we complete this presentation or if you want to ask anything in between these go ahead it's no problem.
01:12:35
So, I'll provide you a checker simply this is an analyzer actually I can show you better probably because
01:12:44
we have already developed this tool so you can have some good ideas about
01:12:49
Let me just open up the site.
01:12:55
Okay, so let's open for you on our website and this is how
01:13:00

the Chrome extension looks like. So you can see we have, I'm actually directly using the demo, rather than I show you, I can go back to the product details and the expectations that we have. So what we have,
01:13:18
You see like a responsive reviews and you can preview your website on full target desktop, you know tea I think you have 27 new goals available here and you can see real-time
01:13:31
You can see that maybe the website has certain issues that generally you have to play right click on the browser is back and then you see the console not.
01:13:41
So you can see those problems here and for example if you want to test your website on real devices then you can directly open the Kaiok device then
01:13:55
face.
01:13:57
and on the real device side you can see currently available online devices basically and then you can directly test you know
01:14:06
device directly you know from this URL device this is how it works so I will just for an example because I
01:14:17
I will just open it so you can see this is how it works. So for example, you just open your
01:14:23
your back plate form that is in the recommended stage right now and you want to do a quick
01:14:28
testing for the responsive design for example
01:14:33
you can just go to the correct station and locate any one of the
01:14:38
you know
01:14:41
how it appears on the laptop or a whiteness top, you know, so the viewport will automatically adapt to that. For example, you want to see all.
01:14:49
how my website appear on a whole trailer.
01:14:53
You know, so it's very easy to press.
01:14:58
any issues that are happening you can monitor them in real time.
01:15:05
and uh...
01:15:08
You can also generate a readiness report.
01:15:14

Just by clicking on it and it can tell you like mobile readiness school verify the redacted risk on physical Android devices
01:15:24
We always want people to go and...
01:15:27
test on the real devices in the device now.
01:15:33
and you can copy the survey or you can download the report. So there are certain features that we already have tested but we still have a lot of
01:15:49
you know, problems, some reports are not being generated correctly or the
01:15:57
user friendly actually. So this is how the kyotro extension look like you can download you can try if you want and
01:16:07
So it will be much easier to understand.
01:16:26
So analyzer we already have this feature in place you know but it's not
01:16:33
and girl.
01:16:36
So we want to improve it.
01:16:40
I think you guys need to be careful.
01:16:48
extension and device that connects the automated browser
01:16:53
you know, scarce to a real android device for example
01:16:58
of the U.R. to the left side.
01:17:00
view polls in the code extension but if you want to if you want to open that
01:17:08
you just click like this on your device and you know the
01:17:14
you know, rear device session and it will open on the rear phone.
01:17:23
Mumbai QA responsive, you know, is not equal to
01:17:29
I mean it will depend like how much you have the editor, the usability, the data station, sometimes we have.
01:17:37
website they have a stop on like okay hamburger menu

01:17:40
not open-aid call different problems you have to identify so the responsible user is simply being responsive does not guarantee a good mobile experience common layout bugs
01:17:51
the usability before users have been interacting with other features like horizontal overflow
01:18:01
any text, smart, the started and over-dapping elements, broken mobile navigation and image scaling.
01:18:10
So different challenges, you know, when it comes to...
01:18:14
and for the usability.
01:18:23
So the primary users
01:18:50
the potential user because you are developing the apps, you are building the web platforms.
01:18:56
Small businesses is cool and not profit organized.
01:19:03
So, you can see that the code key way chat features, viewport chat, viewport validation and
01:19:10
touch in text, touch target, sizing analysis and text readability check. Sometimes the text isn't that weird.
01:19:20
the pen side it's not delivered
01:19:23
element integrity element overlap detection responsive images and basic
01:19:29
Accessibility is something that I really want to put into the scope because
01:19:37
you know we have
01:19:39
We are still trying to do it. We tried something but it didn't work out well.
01:19:46
This is partially implemented in the Chrome extension, but I think we still have to improve this feature by testing it, figuring out the bug and what is there to be improved.
01:20:05
like technically breakdown like layout, accessibility, navigation.
01:20:11
finding length clearly by seniority direct connection between
01:20:17
direction and the visible issues.
01:20:22

So this is basically like the current report looks really bad and you can try it. I think the genetic
01:20:31
Yes, all this bones I think and we are pulling a lot of data there but we won't
01:20:36
have more users than the report even for a charity case for example if you are doing a
01:20:46
using codecs or
01:20:48
so you are getting your best
01:20:54
Again, for core-action person that you can basically, you know,
01:20:59
or log into the device there and you can ask
01:21:04
to test on the real devices in the device lab.
01:21:08
Yeah, so that is something that we are trying to do and being also
01:21:22
to see the readiness report in your website,
01:21:28
be fixed.
01:21:31
Yeah, so this is a very high level, you know.
01:21:34
picture, you know, it doesn't look like, you know, in the group vaccination look very bad actually, but this is what
01:21:40
So this can analyze any webpage for mobile issues, it won't add instant mobile Q&A and then back by all new devices and get the report.
01:21:50
example keyword finding, this is just
01:21:56
like horizontal, over-flow, touchdown, kiss size, you know.
01:22:02
say that you may have one.
01:22:06
at portable QA report.
01:22:10
basically having the test metadata including test
01:22:19
and it's not just we are telling the problem, we are also telling what needs to be fixed and how it needs to be fixed in the program.
01:22:30

device lab call to action so after the user test a responsive on the Chrome extension they can basically directly test all the real devices and
01:22:42
device.
01:22:46
and dealing with it.
01:22:48
accessing real device over the internet and now you can directly invoke your
01:22:57
that you are testing in the Chrome extension on a real device in the device lab.
01:23:05
device connection automatically can act as a first leader to detect potential layout and accessibility issues.
01:23:14
We are passing the URL context to the device that will confirm the behavior.
01:23:20
from the physical hardware.
01:23:23
extension, strategy, recommendation, I mean it's just a recommendation but
01:23:29
up to the students how they want to approach this project you know. Optioning
01:23:37
already have a existing coyote extension so we can we can use it and you know we can build
01:23:45
basically we have also implemented some of them
01:23:53
initially
01:23:55
But we over a period of time like we have
01:24:01
and many of those features may be now in place so it's kinda
01:24:08
already happening you know officially but if
01:24:17
can use it as a reference you know how it was and just look our own so that would be
01:24:25
So, do you just stand alone extension, you know, if you feel that the code base is not mature enough?
01:24:35
Yeah, so browser extension architecture, I'm not the right person to talk about it.
01:24:40
technical distance I'll not that good but I will try my best
01:24:43

So, what would you like to say?
01:24:47
He's taking care of all the tech.
01:24:49
But today he is not available somehow so you know I have to I have to
01:24:55
aspect of
01:24:56
architecture of the UN just like that
01:24:59
So, currently if you move the extension you see a pop-up and then you see different view ports and the devices on the right side.
01:25:11
So we want to add these features like one click you get the report without opening the extension if you want to do that.
01:25:20
Then get this script, inspect the dome and layout properties, background script, coordinate, scale, state and device lab, hands-on.
01:25:33
mobile cubic angle design constraints is you know
01:25:39
inclusive must never break after or modify the tested website.
01:25:46
Yeah, so it's not like the extension is modifying or ordering the website, you know somehow yeah, so it should not happen
01:25:55
Zero noise avoid collecting unnecessary user data and minimize zero
01:26:01
positive, you know, so we need to make sure that what we are carrying is the
01:26:07
about their website, making it sound like that, okay, everything.
01:26:14
good. Clear language explains what fine things clearly in plain actionable language. Right now
01:26:23
The current reports are not like that. It's very technical. It's not easy to understand for a normal user. And we want to simplify that because a lot of people these days, you really know me to be a
01:26:37
having all the technical skills to build something because you can use code, etc.
01:26:42
are there to help you out but though you can build it but you cannot understand technical error if you do not have that background so we want to simplify that part too
01:26:53
Once the data should validate useful checks which are most useful

01:26:59
and developers report for main port format is most practical just on HTML or share thing
01:27:07
Sometimes of globe, how should device that hands of the structure is moving? It's working correctly but it's not really smooth sometimes when we do the hands of the real devices.
01:27:20
Sometimes it works or some devices it doesn't work but I think that it will be ruled
01:27:26
due to the OS limitations or if you have not set up your, you know, for example,
01:27:32
and use some publisher or you know what I mean.
01:27:38
some information like starting in a gas mode or something.
01:27:42
QA checker MVP success criteria instant scan user can run scan in one click and understand the report instantly CBRDRank team issues are ranked CBRDRankX
01:28:00
or table for team sharing, device bag integration. I think I have already covered this part.
01:28:06
So it's just not really the project comparison. I mean it's two different domains. One is healthcare and another is like TAFTU.
01:28:17
The main is for cleaning operations, using clinics, patients and staff. Main risk is that because healthcare is very sensitive data, so privacy and work complexity, we have to
01:28:30
Those are the main risk deals for the Mumbai QH anchor.
01:28:36
I think it's still in a good shape because we have already
01:28:40
up to some
01:28:42
but it's not stable and that is smooth you know we really wanted you as a developer
01:28:50
you are developing web, you know, mobile ads, probably you yourself, you know, the customer for the moment.
01:28:57
you know for the extension so we can we can you know we can interview you know ourselves and ask like what are the challenges what we are facing now and how we can improve that you know by implementing or by supposing a new teacher so I'm like more than open you can also propose
01:29:15
new feature, new idea to improve the extension that like really bring more value to the to the crop extension.
01:29:25

So the users for this is development and QA teams. Yeah, I think I mentioned about the main risk before as well about this. How students should move from
01:29:38
is to discover interview users and clarify constraints.
01:29:44
design, map the workflows and the UI screen.
01:29:50
You can use design, print, you know, maybe inviting group of students, people, some of the people, or students from
01:30:04
from who has medical size, background, or health.
01:30:12
and can basically come up with a
01:30:17
or a storyboard or you know, sub.
01:30:23
used to move forward.
01:30:26
discover design
01:30:31
I mean you can use
01:30:34
whatever
01:30:40
We have to make sure what we are doing is something usable, we can validate the clinics, we can all go to some patients, at least we can, I know because the time is very limited, we cannot do anything.
01:30:56
very good test with users, reports, and real devices, demo shows, the problem,
01:31:08
Yeah, so the fact that the strongest final demo connects user pain points, product diseases, and validation and this.
01:31:16
Final guidance start with users that I think because they are going to use our
01:31:24
start with you start with user
01:31:26
not underlying technology stuff. Keep and read these mods. So I'm very much flexible to the scope. It can adjust based on the time limitations that we have, this limitations is not a problem. So build the smallest useful and we have key privacy and safety in mind. Test on real devices, design for real devices and validate assumption with the real users.
01:31:57
Yeah, I think these are just some suggestions.
01:32:02

Let's get this on.
01:32:06
Yeah, I think that's all I have here. You know, share with you guys. I'm not feeling better.
01:32:12
I got a cold and calm
01:32:15
Please let me know any questions that you may have.
01:32:37
You can hear it? Yes, it's not good.
01:32:51
I have a question about the
01:32:55
exercise so how many staff members accommodate
01:33:00
volume.
01:33:02
Okay.
01:33:04
Okay, yeah great thanks for your question. So right now we have
01:33:10
validated this with the cleaning but I expect like a small
01:33:16
you know and based on the campus city on an average for example
01:33:21
like 10 to 20 patients you know so we expect like
01:33:28
and at least by 20 stop at this point.
01:33:32
You know?
01:33:34
And this is for multiple clinics, right? It's not just for one.
01:33:39
So basically we have multiple patterns.
01:33:47
be supported by English or...
01:33:55
I think French, if we can support French, I think that would be the best we have, considering Canada, because we can also escape to maybe Montreal.
01:34:04
you know, more, you know, finishes poking through it as well.
01:34:08
Thank you.
01:34:25

I was wondering how is this company making money and more specifically
01:34:30
Are you getting this outright as one thief or is it like a subscription based for a month or like a week?
01:34:37
So your question is about the key workflow, right?
01:34:42
Say that again, sir.
01:34:50
Yes.
01:35:09
We want to launch, once we have this MVP, we have to log in.
01:35:16
certain level
01:35:19
and staff and we want to find a founded focus.
01:35:26
So
01:35:28
Because we have to work on the business plan.
01:35:33
is the standard strategy operations and all this stuff. But the way it's going to work, for kids, it will be a subscription plan. They can buy the monthly subscription. And yeah, that's how it's going.
01:35:48
I don't have the right answer because we have not worked on souls.
01:35:52
different laboratories and only we define the place of the lab.
01:35:57
Thank you.
01:36:04
and they'll go up.
01:36:08
Thank you.
01:36:13
I just wanted to ask about
01:36:18
for a clinic though, how would you be screening these appointments?
01:36:25
Let's see.
01:36:27
giving them a list to choose from for like what they're wanting.
01:36:31

for or would you?
01:36:34
that's it because I believe if you start asking for information on my
01:36:40
medical symptoms or anything.
01:36:42
are created now in the EMR system, right?
01:36:49
your family clinic right you just get you the appointment and from your own card you already know who is
01:36:58
just about I think basic appointment is scary, you think better than talking about you know being down about the problem
01:37:06
data that we do not want to touch actually because that requires a lot of
01:37:13
a legal company and we want to end that data. I think we are not prepared for that.
01:37:24
people book appointments
01:37:26
any reason what's stopping them from...
01:37:31
that people mean to, but like wasting doctors' times taking other people's appointments. Lots like...
01:37:38
The clinic is actually being used properly.
01:37:42
So, yeah, this could be a problem. So one thing that we need is, you know, the back portal, for example, is about the patient's isolation, the booking.
01:37:55
We already had in the holy God of
01:38:02
day, you know,
01:38:04
doing you know false appointment
01:38:10
doctors are just trying to understand why they will do it right wasting the doctor time.
01:38:17
Most likely we don't want to go in.
01:38:23
because I
01:38:25
the clinic already knows who are the origin

01:38:29
the health card address with the clinic so the people who are having you know access to that clinic we already know who they are.
01:38:37
I do not understand your perspective.
01:38:46
It was mostly just
01:38:49
Maybe people
01:38:52
don't have an issue they perhaps
01:38:57
do and then overreact, blow it out of proportion and then they go and they book this doctor's timeslot for however long the
01:39:08
Yeah, so
01:39:12
I think
01:39:17
Maybe you can do one thing.
01:39:23
So doctor can speak a bit about this patient and then probably when he tried to book the appointment
01:39:34
thoughts if it doesn't show up I mean to figure it out honestly I do not have the right answer but
01:39:39
I will think about it and maybe also validate this with maybe some basic process. We can do the evaluation and see if this kind of scenario is happening there, right? And we can find out how to handle this situation. Thanks a lot.
01:40:03
Thank you.
01:40:11
I was wondering if you had any plans for some sort of accessibility for example for people who are
01:40:19
or who do not have the best eyesight and if there are audio features we could add on the central system to make these
01:40:30
I think that would be great. I didn't consider them as part of NGC, but I think you're right. I mean it all depends.
01:40:43
because that will be really at this point because before I was working with the daughter.
01:40:51

But they are like, it's more likely, your four students, so I'm not sure how it's gonna work this time.
01:40:57
I'm looking at the theme, you know, people have those challenges and if we get that accessibility features in the application that would be, you know,
01:41:08
Great evaluation.
01:41:14
For clinics, does it make sense to you expect this to be the only sort of forum that they're using? Or do you think that there could still be an option for a client to speak directly with a receptionist or some kind of person like that?
01:41:27
I know a lot of people prefer to have one more in the attraction.
01:41:33
or if they just aren't the best.
01:41:38
So,
01:41:50
So inside is still at the end, maybe the patient then sets the dog.
01:42:02
familiar with how we use the digital system and maybe our own person, maybe having that technical, but I think from the patent point of view, the clinic should still use the clinic flow to create this regard for this application.
01:42:17
Instead, we call it the vapor bomb.
01:42:19
Does that make sense?
01:42:24
Thank you.
01:42:35
My question to you is are there other products like this or competitors that you hold as the gold standard or take inspiration from?
01:42:45
There are so many computers. I like it. So what is the way I look at the completion?
01:42:53
I see the market as a slice.
01:42:58
So you have a cake, you have a market size and all these computers taking a bite of that cake, right? So we are trying to take these from there. You know, by building something more innovative, solving the real problem. Yeah, and so yeah, I mean, there are computers.
01:43:18
Honestly, I didn't have all the
01:43:23
and all the data, but I know two companies.

01:43:27
And I'm trying to
01:43:32
I'm trying to look for a founder somebody who can work with me together, you know on the business side of it
01:43:39
has been needed, business plan, muggling strategy,
01:43:45
So
01:43:57
and we can do the count as your analysis check out the code.
01:44:04
We have a very limited resources, but yeah, I'm more than happy to work together to figure it out from the market size perspective.
01:44:13
for the public perspective. Thanks.
01:44:16
Thank you, sir.
01:44:18
Thank you.
01:44:25
How's it going? So I know you said that you didn't want
01:44:33
too much in depth with patient profile, like you don't want to die.
01:44:37
or
01:44:40
notes or whatever.
01:44:42
But would you consider having a patient belong to a certain clinic or have a patient with a family doctor or something like that or a running
01:44:55
like tap of like injury or illness.
01:45:00
their profile.
01:45:07
Right now we just want to have
01:45:13
and clever interaction.
01:45:17
You know the sensitive data about your particular sickness or problem
01:45:22
not want to

01:45:28
for some reason there are no
01:45:32
Even on the device that we are having a lot of security challenges and I think I'm very much worried about the patients in the personal health report data that needs to be really protected.
01:45:45
really planning to put it in this scope until we know how to handle it.
01:45:50
So yeah, we are just I think keeping the interactions very much limited between
01:46:02
and the patient and the doctor, you know. Most, more or less is like,
01:46:10
just getting in, just getting in tension and sharing, you know, just those what goes.
01:46:15
And yes, we made a
01:46:21
visit or find out something later on if it doesn't work. I'm more than happy to hear from you how we can improve it or how we can do the validation with play eggs and figure out the real-time point and what level of data we should touch based on our capabilities. So I'm more open to hear your feedback, your ideas, your collaborators, and you can find out how we
01:46:51
But really he will not want to touch any end.
01:46:56
data of
01:46:58
and then the broken, yeah, practice board.
01:47:01
Thank you.
01:47:03
some.
01:47:16
So I was wondering if your app would be able to show
01:47:21
because if you want the clinics and you have
01:47:25
patients. Would clinics be able to show if they have openings within the clinic for new patients, so people who are
01:47:33
Canada and the provinces are also people who are unable to find
01:47:41
to go to for like long term to join a clinic and I was wondering if there would be the opportunity to allow clinics to show that they have vacancies available to take on new patients and whether people who want to join a clinic
01:47:55

associated with the clinic, get on the app and be able to make that connection.
01:48:02
Thanks for the question. I think this is also the very basic process in the plan. Immigrants move into Canada and they have to find out which clinics are accepting the new patients. So yes, we will allow this as a process.
01:48:30
the new patients because it's very important otherwise we will just keep going or keep trying to take it off. So yeah I think we have to have this availability.
01:48:40
which clinic are accepting the new patients, yes, you will allow that.
01:48:45
Thank you.
01:48:52
Hello sir, I had a couple of questions regarding the Jigarom hilarious app that's intended for
01:48:59
So this is not going to be province and region specific. For example, it has different regions for different provinces regarding their first province.
01:49:12
Right now we are very much focused and we have to speed in Ontario, here on this call, and even in Ontario. I think I have a plan to work with people in the GBA area because I am based in Mississauga.
01:49:26
And if I find some potential student, because I'm a
01:49:31
I think
01:49:34
some students that you want to try to figure it out, the students who have like medical science background, you know,
01:49:43
of the system, if I find potential person to work with maybe with our
01:49:49
also
01:49:56
I might
01:49:58
in Mississippi and Jordan, almost likely.
01:50:02
Are you thinking of expanding countrywide?
01:50:06
Yeah, that's the scaling plan and that depends like
01:50:13
We have heard how
01:50:18
you know in the real world you know

01:50:22
and stuff if you are really solving you know there are pain points or challenges but if you see that yeah we are making good traction in exciting you know people find it
01:50:34
to scale and to scale into the province and we may need to do some
01:50:41
based on, you know...
01:50:43
and links are also in the same required.
01:50:48
Would you target go-by clinics in North Canada as well?
01:50:52
because people in that area couldn't know that the clinic was coming by to run.
01:51:01
Honestly, I didn't...
01:51:05
I think that's fine yet, but I'm okay.
01:51:08
you learn more but yeah we can put it into this
01:51:13
and also
01:51:15
challenges and I think it's all about what we learn from the patient data current challenges clearly.
01:51:24
Yeah, we will put that into scope.
01:51:29
I wouldn't be able to link multiple financial hospitals per day.
01:51:33
doctor's review specialist
01:51:39
you have a we're all
01:51:41
Thank you.
01:51:43
We just finish and then we will be in the scope of this movement for cleaning.
01:51:50
hospitals at this point so I believe like I'm not sure if a person can
01:51:57
one family doctor assigned to it, if that is the scenario, so wherever your register you should be able to see the appointment.
01:52:06
My question was if the family doctor writes you to a specialist, would the appointment of the specialist also be on the app?

01:52:16
Sorry, thanks, and I want to understand your question.
01:52:22
Yeah, actually the specialist on the app
01:52:28
I mean it could only be possible if they are also part of a little family cleaning and you know and if they are registered on our system if not I think
01:52:41
I mean I honestly don't have the right answer for this
01:52:47
how it needs to work clearly. But I think right on the current
01:52:52
and the doctors at the clinic. If they are there in the system, it should be, you know,
01:52:59
doctor and then refer you know for the session I specially
01:53:03
Okay, just a few more questions.
01:53:07
to look into stores, personal identification number, and storage login data, or are you going to use things like
01:53:15
No, email address, you don't get it.
01:53:20
that's the only thing we want to use for login or maybe you can add your biometric, you know, fingerprint or faces can face ID. You have to login. That's the only thing I think is go back to site.
01:53:33
Thank you.
01:53:35
Thank you.
01:53:46
Hello, sir. I just wanted to ask, have you considered how piloting flow could accommodate international students or newcomers who don't have the O'Hare card, especially when it comes to booking appointments?
01:54:04
Yes, so I think this is also
01:54:08
So, because the paradigmic register that coming you know.
01:54:15
the link is coming to Canada. They also want to have a whole coverage site. That's very important. So yeah, we do support it. And like, you know, I
01:54:27
So for example, you are based in Mississauga area and the woman she put a fill code of your area, you know, you can see the clinics.
01:54:37

you and you can see where they are accepting the new patients for example and you can schedule, you can make an appointment with that and you know have a
01:54:47
The family doctor beside is also the main thing that gets the process.
01:55:09
about access to services that aren't always open to everyone such as like your family doctor and just kind of we're wondering
01:55:18
there would be a process kind of to like be able to find those workings with your family doctor instead of
01:55:23
everyone being able to see it, it's not public.
01:55:26
So basically your data is secure.
01:55:31
So you'll find me.
01:55:34
you know the feeling accepting you patients or not just only public information you know so you know like if you then if you can you know
01:55:48
doctor at clinic because they are accepting your appointments but not your personal appointments.
01:55:56
We are not going to show that.
01:55:59
So sorry, one more. Would there be a process, would you have to go through your clinic then to be uploaded?
01:56:07
like be able to have access to those appointments that aren't available for
01:56:14
if you go to your family doctor.
01:56:20
Maybe I still have a little heart I've got is that can you please repeat the game?
01:56:29
family doctor not everyone can go to it's your family doctor
01:56:35
In order to be connected to your family doctor and see what their little pocuelos that are, would you have a process to
01:56:43
to get there to be raised to your family's office.
01:56:47
Actually the process because I think this process is already in place with the
01:56:58

This is the doctors list of labels and you can choose, right? And you can see what is there. I think that is a complicated question. If you go in general, I see the
01:57:09
They already have the doctors that are working on that gaining their specialization and generally they ask you to choose.
01:57:18
doctor is accepting the new patient and the TV was just approved.
01:57:27
is not the doctors.
01:57:28
schedule the appointment. That's how I think it needs to work. Probably this process is already
01:57:35
Thank you.
01:57:45
Thank you.
01:57:52
Hi, Avi. So I just wanted to know how should users initiate the one-click analysis for the QA chapter? Like, do you see this tool being built as a web application, a browser extension, a command line interface? Like, what do you have in mind for it?
01:58:07
So like I mentioned we have built it as a Chrome extension right now and also we have a bad person of it but there are some limitations related to that.
01:58:20
and sub-webside that do not support it to the environment. So it's not working very well.
01:58:28
So right now we have like Covid station and the
01:58:32
If you go and sign up on the device
01:58:36
you a menu in the lamp and you can do the same what you do with that center but extension has a lot more
01:58:43
features and yeah, a long session forward.
01:58:49
Yeah, so the board is...
01:58:51
How is working but it's not working.
01:58:54
the text is very well on the vertical screen size.
01:58:59
If you want to automate that process and currently we are generating some reports on this program based on
01:59:07

But it's still not that good. So we want to improve it even you define like we really need to figure it out.
01:59:17
report to the
01:59:21
Thank you very much.
01:59:24
I think that users are simply experienced.
01:59:27
Yeah, so the idea is to continue.
01:59:32
from extension or the
01:59:37
But until then, if you have like some
01:59:41
some good reasons why we should have a problem lying by your side.
01:59:46
Right now I think it's a Chrome expansion and that won't be a bad thing.
01:59:53
at this point. So please let me know if I have tested your ocean correct?
02:00:00
Yeah, no, that's great.
02:00:03
in question really quick. For the QA score weighting, do you have an existing formula or like a preference for calculating the overall QA score or should we propose a custom weighted algorithm for that?
02:00:15
So we have something already in place. I think my senior year is in South Virginia.
02:00:23
I can share what is currently on the labor and how it is being calculated and definitely we are very much open to
02:00:35
Hello, your voice.
02:00:41
We can adapt to this. If you have better ways of doing it, just please be prepared for those.
02:00:46
All right, thank you so much.
02:01:03
Would you like that pair of ID scanners so that you can confirm who ever is looking to the appointment?
02:01:13
So we want to do that only on the
02:01:18
And at this point, it was a

02:01:27
And even the bio-matics

Summary:
What Clinic Flow is supposed to do

For patients:

Find clinics, potentially by location
See which clinics are accepting new patients
Book appointments
Manage/cancel appointments
Get automated reminders
See waitlist status
Potentially access appointments with their family doctor
Support people without an Ontario health card, such as newcomers/international students
Eventually support English/French
Have accessible options for users with disabilities

For clinic staff:

Manage appointments and schedules
Manage patient bookings
Manage waitlists
See clinic/provider availability
Handle cancellations and no-shows
Manage multiple clinics/locations
Have a dashboard showing things like:
Appointment volume
Volume by provider
Cancellation rate
No-show rate
Average waiting time
Waitlist fulfillment speed
Provider utilization
Overall facility utilization

Meeting Transcript

And here we have I'm Juliana Sangregorio I'm Christopher Wee I'm Cheyenne Qureshi and I'm lex Moore i'm over here and we'll be prevent presenting our project overview for clinic flow Just one thing what is your team name Team Equilibrium OK sorry equal equilibrium That's in in balance OK yeah So for the meeting today we want to present our understanding of the problem and review the draft problem statement that we've come up with propose the designed concept and discuss target specs and constraints and after hopefully gather some feedback in the bottom of the slides there are some questions although I know that we don't have too too much time If we do have more time we've come back to OK so our problem statement so we know that small clinics in Canada have difficulty managing appointments communicating with patients and just keeping track of their schedule and these tasks can lead to a lot of missed appointments a lot of scheduling issues and just a lot of miscommunication between patients and clinics so there is a need for an affordable system that will help clinics manage appointments send reminders and again communicate easily with the patients And basically the goal of Clinkflow is to equate like a simple platform that will improve important management and will be able to communicate with patients Yeah the requirements that we decided for clinic flow was that it should take less than five steps to do most tasks it should have less than a minute of notification speed so whenever the clinic would release information about your appointment it would take less than one minute for you to receive it on your phone It should be up 24/7 and it shouldn't have any major maintenance stops and should be available for everyone to use at all times of the day We should

have 100% appointment detail so when the clinic is releasing any data or we show any data about your appointment you should know all of it should be true and you wouldn't have any miss calendar dates or anything like that it shouldn't fully comply with all of the data standards of our area so that would be Ontario and Canada Our constraints should we should follow proper data security and have audit logging AUI should respond in under A second and our operating cost should be under $50 a month So our idea has five band subsystems a global UI a database and authentication for storage and security data accessibility settings that allow our system to be more inclusive a notification system that keeps our patients informed and notified and a clinic Finder and appointment log So for our proposed prototype we have two main prototypes that we were hoping you could help us like find which one you prefer yeah so feel free to just let us know if you would change anything or if you like either one Yeah I'm just trying to understand UX so in the appointment management for example if the user want to cancel an appointment or modify an appointment uh why they are like two different buttons you know probably uh either you can modify or cancel i'm just thinking uh how how you came up with this idea yes I'm just thinking from the end users perspective you know if I will get confused or this is how it should work you know so yeah just just a just a general question you know I I'm sure you guys may have some brainstorming to come up with these ideas Yeah so for modify appointment it was more like you if you want to change the date and cancel is really just I guess cancel the whole appointment but we could always like redesign it so that it makes more sense so it's one button instead of 2 Yeah I'm just thinking like for example you click cancel an appointment then you are going to see the list of appointments and then clicking a cancel button there further or how it works yeah same for the modify so again I click on modify then I see tense of appointments there you know scheduled for for a day you know whatever and then oh OK understood sorry this is

for the patient Yes OK it's perfectly alright so there is only one appointment sorry II misunderstood that part OK now it's clear OK oh OK OK is this perfectly all right sorry I misunderstood no worries thank you Alright so this is our global UI like another prototype design our global UI's main goal was to reduce the amount of text on screen to deletter the home page essentially decluttering the the home page allowed the patients to have an easier time finding private functions such as booking appointment information and clinic announcements without a secondary action Secondarily it includes all the accessibility settings like language contact information and FAQ And then of course there's also a UI text contrast to improve the visibility of all the functions As for the database and authentication we want to have secure data transmission using transport layer security HTTPS so when a user creates an account on clinic flow and they are provided with an API authentication token we can figure out the exact method but I've come up with using either Oauth 2 or Open ID Lots of different ways that we can go about about that I did want to get your feedback on what database you guys are currently using or planning to use is it'll give us a much much greater communication of the direction yeah so we are using Firebase I mean you can if you want to use Superbase also that is fine we also use Superbase sometime but yeah Firebase and we I can tell you more precisely just a minute because I'm not really good on the tech side my CTO take care of those stuff OK if is there an email that we can yeah it's it's a firefighter store Just what we are using OK Thank you yeah and yeah we just have different Jason tables to store the data next is accessibility settings so here's just a list of different accessibility settings that could be used I would like your opinion if you think these are good if you think we might not need some of these or if there are any others that you would like to include No this looks pretty good actually very well thought so I'm sure there might be some patients who really need these settings they may have some kind of disability so

it's pretty good Thank you Yep so this is our notification system the notification system follows an algorithm that helps it decide what it sends how it sends and when it sends it relies on a primary primary push notification system and then a secondary fall back system just in case that push back system fails Umm so the secondary system essentially just sends an email to the patients and essentially all of this keeps a log of all the notifications that were sent out to the patients advert appointment logic and clinic Finder our system allows patients to manage their appointments by booking rescheduling or canceling them it also helps users find nearby clinics based on their preferred distance they could view clinic addresses and filter clinics by appointment availability So the number one that we wanted with that is to have it in under five clicks for the user for almost every step does that sound like something that's reasonable for clinical Yeah it looks pretty good and first of all I'm sorry because I misunderstood the first I was like I was a little lost and then I realized sorry I lose the context but yeah II think this is pretty good and very well thought system just one question how you came up with the pricing $50 have you done any research or any data like how how will you back that data for the pricing we were we were unsure wanted to ask your business model that we said less than $50 a month to give leeway if there's any subscriptions required we did look at Jane up and tell US Health but as for pricing we're a little up in the air right now with with what your business model will look like OK have you guys prepared any kind of business model canvas or something which I can see it's not the most important thing i'm just trying to understand because I think you guys really thought deeply about it so II was expecting maybe you might have done some user research or some data maybe you might have collected because I can see some some of the stuff reflecting in your presentation we do have a tiny tiny bit but not not enough to give you precise details we wanted your opinion on what funding looked like and how much you guys were planning on on charging

if that had been thought of if not then we will go ahead and spearhead that Yeah so from our side like this could be a new company in the future OK if if you're able to really build it to that extent you know that we find some clinics who really want to use it right we can use our team doing demos and see the viability you know from the clinic if they really like the idea and they want to use it so it could be a company in the future and yeah we may think about it but if you guys need you know any help on the development side for example you need 5 best project access or anything I can help you with the cloud subscription so I can give you the Blaze plan you can you can leverage that if you need a cloud account I can I can I can help you with that but I think if you guys want to bring it really you know this product into life and if you can do a bit more you know use the research you know work with few clinics collect some data understand their pain points understand the patient's pain point and see if if they will use the system awesome and and if you can think also about the integration like what are the current system how they are managing currently you know and the potential integration because that's how you penetrate the market so So that's what you know I think that that I can suggest you probably a bit of more research you can do think you own clinic flow if you are the CEO how will you decide I understand now as a business thank you so much Abby we do have a little more time unfortunately but I appreciate all your time today and your comments thank you so much thank you thank you everyone

GNG1103

Project Deliverable B

Equilibrium (Team 5)



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2026-10-04

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Table of Contents

1 Introduction....1
1.1 Problem statement....2
2 Empathy....3
2.1 Empathy process....3
2.2 Raw user data - Shyaan - https://forms.gle/trnSEiFtZ85QrcuNA....4
3 Needs and Design Criteria....5
3.1 Organization of Needs....6
3.1 Design Criteria Description....7
4 Technical Benchmarking - Christopher....9
4.1 Product 1....10
4.2 Product 2....10
4.3 Product 3....11
5 Target Specifications....12
5.1 Functional-Requirements....12
5.2 Non-Functional Requirements....12
5.3 Constraints....13
6 Conclusions....15
7 Bibliography....16

1 Introduction

Canadian clinics and patients are currently relying on a fragmented administrative system in order to keep track of all their operations. Existing technologies are too bulky and complex, leading to many inefficiencies and bottlenecks within their operations. This creates a frustrating and tedious process for the patients. With costs rising, smaller local clinics suffer from high overhead costs [1], while complex billing software can drive up operational expenditure without improving patient care [2]. This makes a proper connection between patients and clinics much more difficult. This document is organized into six different sections with a reference section at the end.

The first section provides insight into the problem as well as any assumptions that were made during the initial brainstorming process. The second section is the empathy section; this section describes the process our group used to collect information and the information that we have gathered from users. The third section focuses on the interpretation of the data that we have collected; this includes information regarding potential needs and the criteria that we developed regarding our design. Section 4 revolves around our established competitors in the industry; this provides information regarding features that our competitors use and how we can incorporate similar types of designs, as well as features that are not needed. Section 5 is focused on the design section of our app; it will go into further detail of the features that we had researched in the previous section, as well as a description of how necessary that feature is to the system. It will also provide the constraints that our team decided were necessary based on the responses of interviewees. The sixth section provides the reader with a conclusion generated by our group; it

will describe the main functions that our group intends to add to the software as well as any final impressions or insights that our group believes is important.

Our group's main assumptions surround the stakeholders affected by our solution as we assumed that the main stakeholders will be clinics, patients and doctors.

This deliverable covers user research, needs and design criteria, technical benchmarking and target specifications to develop a system that is lightweight and more streamlined compared to present solutions. By unifying patient scheduling with core administrative data management, clinics will no longer have to rely on the fragmented systems of the past.

1.1 Problem statement

Current software for clinics has become far too bulky and complex, as clinics have to keep track of all their patients, their records, their bookings, announcements from clinic to patient, as well as messages between doctors and their patients through multiple different software. By creating a streamlined software that combines both sides, it will create a more enjoyable experience for the patients while also decreasing overhead costs.

2 Empathy

2.1 Empathy process

Team Equilibrium started the empathy process by reviewing the ClinicFlow project brief and client Q&A to identify the main groups affected by the appointment system. The team categorized these users into patients, clinic staff, and clinic administrators. Doctors are also affected by scheduling, but the project scope is strictly limited to administrative tasks, not replacing existing clinical systems.

For patients, the system needs to handle finding clinics, checking if they are accepting new patients, and booking appointments. The client Q&A pointed out several accessibility factors to consider, including support for non-OHIP patients, different levels of digital literacy, and bilingual or low-vision interfaces. These requirements led the team to focus on simple scheduling, clear waitlist rules, mobile access, and data privacy.

To test these assumptions against real user behaviour, the team deployed a questionnaire. Instead of guessing where the bottlenecks are, the questionnaire asks respondents about how often they book, what interface issues they face, and what personal information they are comfortable sharing. This ensures the design targets confirmed functional problems rather than assumed ones.

Front-desk staff handle daily patient flow and need tools for scheduling, managing waitlists, and checking provider availability. Since cancellations and no-shows lower clinic efficiency, the system needs to reduce manual work using automated notifications and clear availability tracking. The client also specified that clinic setup must take less than an hour and require no manual data re-entry, meaning the software needs to integrate into existing workflows easily.

Clinic owners and administrators need summary reports to track performance without having to look at every single appointment. Important metrics include appointment volume, cancellation and no-show rates, waitlist speed, and provider utilization.

Across all user groups, data privacy is the main technical constraint. ClinicFlow is an administrative tool, not an Electronic Medical Record (EMR). It must only collect the minimum data needed for booking, protected by standard authentication, encryption, and access controls.

The initial client Q&A raised several assumptions that still need field validation, such as how to handle non-OHIP patients and which accessibility features to prioritize. The next step in the empathy process is to combine the questionnaire results with direct interviews, including patients who need digital assistance and clinic staff. By comparing the client's intended workflow to real user feedback, the team will establish the final technical requirements for the ClinicFlow design.

2.2 Raw user data

Questions - https://forms.gle/trnSEiFtZ85QrcuNA

Notes & Observations:

What is the most difficult part of booking appointments?

The biggest problem people had was finding an appointment that actually works for them. People mentioned “not enough availability,” “finding a good day,” and “getting a good time.” This shows that patients need a simple way to see available appointments and choose one that fits their schedule.

Waiting and cancellations

Everyone who responded said they have had to wait a long time at a clinic. One person said it "just wastes time," while another said it made their experience much worse. People also mentioned having to call the clinic to cancel or reschedule. This shows that patients would benefit from being able to manage their appointments themselves and get updates when there are delays or changes.

Waitlists and notifications

Most people preferred getting a text when an earlier appointment became available. One person suggested that the system could know when they are free and notify them when a matching appointment opens up. This could make waitlists more useful and help clinics fill cancelled appointments faster.

Privacy

People were comfortable giving basic information like their name, phone number, and email, but did not want to provide things like medical records or banking information. This supports keeping the system focused on booking and administration instead of collecting unnecessary medical information.

Overall Observations

The main things I noticed were limited appointment availability, long wait times, and having to call clinics for simple changes. The responses show that ClinicFlow should focus on making

appointments easy to book and manage, sending useful notifications, and only collecting the information that is actually needed.

3 Needs and Design Criteria

In Table 1, the user needs are organized around ClinicFlow's main goals: appointment management, patient flow, security, communication, usability, and clinic administration. Each need was translated into a design criterion that can later be used to compare possible design concepts. The priority-based needs are ranked from most to least important based on their impact on ClinicFlow's overall functionality, usability, and purpose.

Table 1. Needs and Design Criteria

Importance	Need	Design Criteria
1	Appointments can be easily managed.	Users can create, edit, cancel, reschedule, and view their appointments in a simple calendar.
2	Patient data must be secure. non-functional	Patient data must be protected through secure login, encryption, and role-based access.
3	Patients and staff receive important notifications.(Subsystem)	The system must send appointment reminders and notifications for cancellations, waitlist opportunities, and clinic announcements to both the users and the staff.
4	The system is easy for clinic staff to use.	Common tasks should be completed in a small number of steps, and the interface

		should use clear labels and simple navigation.
5	Different users have appropriate access.(function)	The system must provide different access levels for patients, staff, and administrators so users can only access information and features appropriate to their role.
6	Important system activity can be tracked.(Subsystem)	The system must maintain a log of important actions, including appointment changes, check-ins, and changes to patient information.
7	The system is accessible on mobile devices.(Different Aspect ratio)	Patients and staff must be able to access the required features through mobile applications.
8	The system is affordable and easy to implement.(Can't really design)	The system should require minimal setup and have a low operating cost compared with larger enterprise clinic management systems.

3.1 Organization of Needs

The needs were sorted into three main categories: patient needs, clinic staff needs, and system needs. The patient needs include easy appointment management, receiving important notifications, mobile accessibility, and having their personal information protected. Clinic staff

needs include easy-to-use appointment management, notifications, and access to information and features appropriate to their role. System needs include secure login, encryption, role-based access, and activity tracking to ensure that patient information and important system actions are properly protected and recorded.

The needs were ranked based on their importance to the main users of ClinicFlow and the overall purpose of the system. Needs that directly affect the core functions of the system, such as appointment management, security, and communication, were given higher priority. Other needs, such as mobile accessibility and affordability, were ranked lower because they support the overall usability and implementation of the system but are less essential to its basic operation.

3.1 Design Criteria Description

The design criteria were created by turning each user need into something that can be used to evaluate the ClinicFlow design. The main functional requirements are that the system can manage appointments, support cancellations and waitlists, allow patients to check in, provide notifications to patients and staff, and track important system activity. The main non-functional requirements are that the system is secure, easy to use, reliable, accessible on mobile devices, and affordable for smaller clinics. The system must also maintain audit logs and provide different access levels for patients, staff, and administrators.

Where possible, the design criteria can be measured using specific values. These could include the number of steps required to complete a task (steps), notification delivery time (minutes), system availability (%), number of supported user roles (number), setup time (hours), and monthly cost ($/month). Security requirements such as encryption, authentication, and audit logging can also be evaluated by checking whether the required security features are

implemented. Since ClinicFlow is still in the early stages of development, exact target values for some criteria have not yet been established.

4 Technical Benchmarking - Christopher

Table 2. Specifications for Benchmarked Products

Product Criteria	Telus Connect	CliniqApps	Jane
Cost User: Provider:	Free to use, with virtual services fees depending on service Free with paid add-ons	Free for patients Payment varies between apps used with services.	Free for patients $54/month Balance, $79/month practice, $99/month Thrive, +$15 for AI/Practitioner
Setup time User: Provider:	Quick app download and connection to provider Quick integration with the TELUS system.	Simple online access “Easy Setup” with integration with existing management system	Web service 5-10 minute setup with additional time for management.
AI use:	Limited AI use, add-on using Tali AI for documents	All tasks are done with AI	AI scribe with paid system. Clinical notes

		Receptionist, communications, clinical notes, task management, waitlist, ect	from recorded/dictated sessions.
Encryption and security	Uses TELUS infrastructure AWS/GCP	AES-256 encryption with automatic data back-up.	256-bit encryption with PIPEDA, HIPPA and GDPR compliance.

4.1 Product 1

TELUS Health

TELUS Health product page https://www.telus.com/en/health

TELUS Health allows for easy communication between patients and healthcare providers [3]. Patients can book and manage appointments while getting reminders and completing questionnaires and surveys. They have completely encrypted messaging and can attend virtual visits.

4.2 Product 2

CliniqApps

CliniqApps product page https://www.cliniqapps.com/

CliniqApps gives almost complete AI automation, reducing administrative work and allowing for improvements in patient communication [5]. Their AI can take messages, phone calls and send reminders and follow-ups. They can work alongside practitioners by creating clinical notes and integrating with any pre-existing practice management system.

4.3 Product 3

Jane

Jane online booking product page: https://jane.app/features/online-booking

Jane's online booking page describes real-time booking, schedule controls by service, provider, location, and time, as well as reminders and waitlists that can offer newly available openings to waitlisted patients [4]. These features address the ClinicFlow needs for self-service scheduling, fewer administrative calls, reminders, and filling cancellations. The page also describes Jane as part of a broader practice-management/EMR product, so it offers a reference for integrated workflows while ClinicFlow's stated concept remains a lighter administrative companion to an existing EMR.

For ClinicFlow, the useful design lesson is to connect a cancellation to a clear next action: identify eligible waitlisted patients, notify them, and update the schedule when a slot is accepted. ClinicFlow should still test whether small clinics need automatic offers, staff approval before sending, or both. The public Jane page does not establish whether its workflow fits ClinicFlow's minimum-data boundary or the needs of Ontario clinics using other EMRs.

5 Target Specifications - ALL

5.1 Functional-Requirements

Table 3 shows the target specifications for the functional requirements of ClinicFlow. These requirements describe what the system must be able to do, such as managing appointments, sending notifications, and supporting different user roles. Each requirement has a target value and a method for checking if it is met.

Table 3.0 Target Specifications for Functional Requirements

Criteria	**Relation (=. < or >)**	**Value**	**Unit**	**Verification Method**
Book an appointment	<	5	Steps	Usability testing
View appointments	<	3	Steps	Usability testing
Cancel and reschedule appointments	<	5	Steps	Usability testing
Display accurate appointment details	=	100	%	Functional testing
Notification delivery time	<	1	Minute	Controlled testing
Supported user roles	=	3	Roles	Access control testing
Staff assisted booking	<	5	Steps	Usability testing

5.2 Non-Functional Requirements

Table 4 shows the target specifications for the non-functional requirements of ClinicFlow. These requirements describe how well the system should work, including its reliability, speed, security, mobile access, language support, setup time, and data protection. Each requirement has a target value and a method for checking if it is met.

Table 3.1 Target Specifications for Non-Functional Requirements

Criteria	**Relation (=. < or >)**	**Value**	**Unit**	**Verification Method**
Uptime	>	99	%	Reliability monitoring
UI response time	<	1	second	Performance testing
Blocked unauthorized access	>	99.99	%	Security testing
User mobile functions	=	100	%	Mobile testing
Supported languages	=	2	Languages	Inspection
Clinic setup time	<	1	Hour	Timed setup test
Data encrypted	=	100	%	Security testing
Audit logging	=	100	%	Log inspection

5.3 Constraints

Table 5 shows the target specifications for the constraints of ClinicFlow. These constraints describe the limits and conditions that the system must follow, including cost, mobile platforms, network access, data use, and privacy requirements. Each constraint includes a target value and a method for checking if it is met.

Table 3.2 Target Specifications for the Constraints

Criteria	**Relation (=. < or >)**	**Value**	**Unit**	**Verification Method**
Maximum operating cost	<	$	$50/month	Cost analysis
Mobile OS compatibility	=	iOS, Andriod	Platform	Compatibility testing
Network connection requirement	=	required	Condition	Network testing
Patient data	=	Not permitted	Condition	Data inspection
Personal data	=	Solely required	Condition	Data collection inspection
Compliance with applicable Ontario privacy requirements	=	Required	Condition	Security testing

6 Conclusions

Deliverable B established the main user needs, design criteria, problem definition, technical benchmarks, and preliminary target specifications for ClinicFlow. The research identified appointment management, cancellations, waitlists, notifications, patient check-in, security, usability, and accessibility as key areas for the proposed system.

Technical benchmarking of TELUS Health, CliniqApps, and Jane showed that many of these functions already exist in current healthcare platforms. This suggests that ClinicFlow should focus on providing a simpler and more streamlined administrative solution rather than replacing a complete EMR or practice-management system.

The project is currently at the definition stage. The design direction is to develop a lightweight, mobile-accessible appointment and clinic administration platform that simplifies scheduling, cancellations, waitlists, notifications, and check-in while minimizing unnecessary patient data collection. The remaining work is to validate this direction through additional user research, refine the target specifications, and develop and test design concepts against the established requirements and constraints.

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GNG1103

Project Deliverable C

Equilibrium (B01.5)



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Table of Contents

Table of Contents. ii

1 Introduction. 1

2 Target Specifications. 2

2.1 Functional-Requirements. 2

2.2 Non-Functional Requirements. 2

2.3 Constraints. 3

3 Global Concepts. 4

3.1 Team Member 1 Concept 4

3.2 Team Member 2 Concept 4

3.3 Team Member 3 Concept 4

3.4 Team Member 4 Concept 4

3.5 Team Member 5 Concept 4

3.6 Analysis, Evaluation and Selection. 4

4 Subsystems. 6

4.1 Subsystem 1. 6

5 Subsystem Concepts. 7

5.1 Team Member 1 Concepts. 7

5.2 Team Member 2 Concepts. 7

5.3 Team Member 3 Concepts. 7

5.4 Team Member 4 Concepts. 7

5.5 Team Member 5 Concepts. 7

5.6 Analysis, Evaluation and Selection. 7

6 Conclusions. 9

7 Bibliography. 10

1 Introduction

Current software for clinics has become far too bulky and complex, as clinics have to keep track of patients, their records, bookings, announcements, and messages between doctors and patients through multiple different programs. By creating a streamlined software that combines these functions, ClinicFlow aims to make the experience easier for patients and clinic staff while also reducing overhead costs. The app will allow patients to find clinics, book and manage appointments, receive notifications, and adjust accessibility settings, while clinic staff can keep appointment schedules updated. ClinicFlow will focus on clinic operations and will not replace an electronic medical record system.

2 Target Specifications

2.1 Functional-Requirements

The following table presents the target specifications for ClinicFlow's functional requirements. It describes how each requirement will be tested, the methods and resources needed, the purpose of each test, and the estimated time required. These tests will help ensure that the app works properly and meets the needs of patients and clinic staff.

Table 1. Target Specifications for Functional Requirements

Criteria	**Relation (=. < or >)**	**Value**	**Unit**	**Verification Method**
Book an appointment	<	5	Steps	Usability testing
View appointments	<	3	Steps	Usability testing
Cancel and reschedule appointments	<	5	Steps	Usability testing
Display accurate appointment details	=	100	%	Functional testing
Notification delivery time	<	1	Minute	Controlled testing
Supported user roles	=	3	Roles	Access control testing
Staff assisted booking	<	5	Steps	Usability testing

2.2 Non-Functional Requirements

The following table presents the target specifications for ClinicFlow's non-functional requirements. It explains how the app's performance, security, reliability, and ease of use will be tested, including the methods and resources needed, the purpose of each test, and the estimated time required. These tests will help ensure that the app is reliable, secure, and easy to use.

Table 2. Target Specifications for Non-Functional Requirements

Criteria	**Relation (=. < or >)**	**Value**	**Unit**	**Verification Method**
Uptime	>	99	%	Reliability monitoring
UI response time	<	1	second	Performance testing
Blocked unauthorized access	=	100	%	Security testing
User mobile functions	=	100	%	Mobile testing
Supported languages	=	2	Languages	Inspection
Clinic setup time	<	1	Hour	Timed setup test
User setup time	<	5	mins	Timed new user setup testing
Data encrypted	=	100	%	Security testing
Audit logging	=	100	%	Log inspection

2.3 Constraints

The following table presents the main constraints for ClinicFlow, including operating costs, mobile compatibility, network requirements, and data privacy. Each constraint will be checked using the verification method listed in the table to ensure the app meets these requirements.

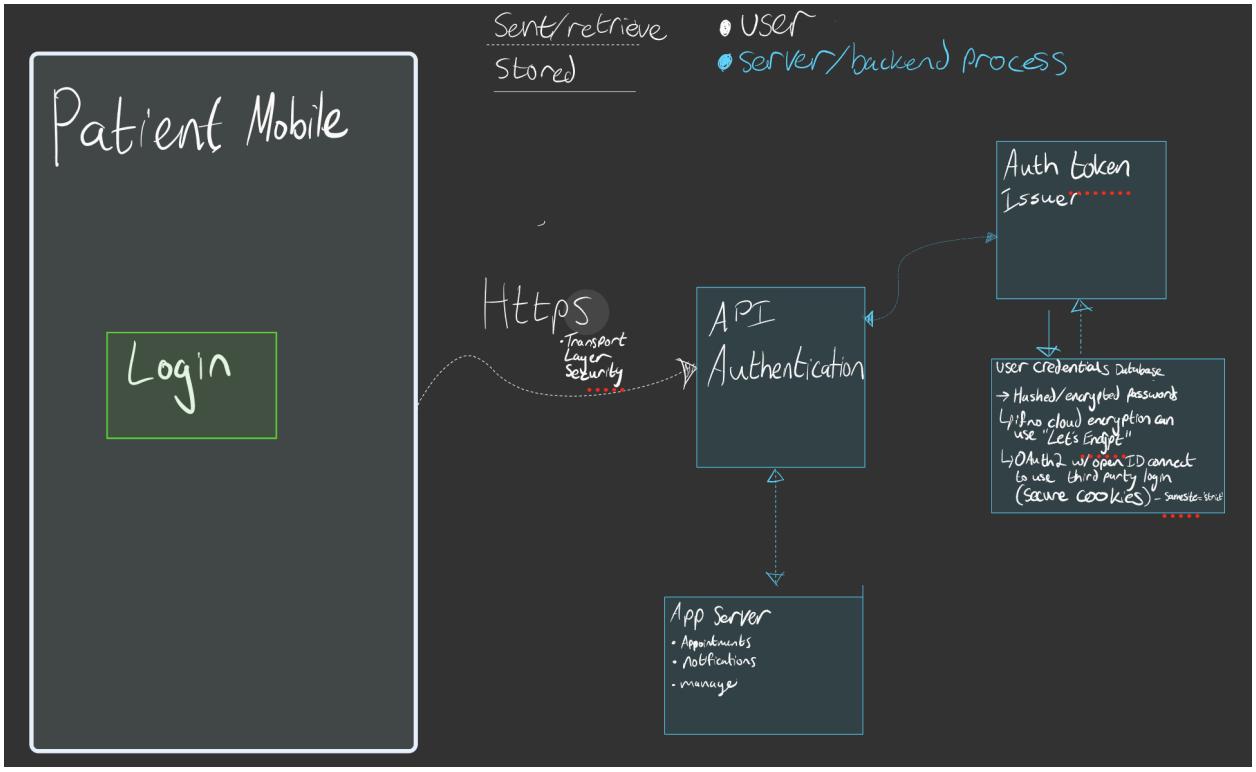
Table 3. Target Specifications for the Constraints

Criteria	**Relation (=. < or >)**	**Value**	**Unit**	**Verification Method**
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Patient data	=	Not permitted	Condition	Data inspection
Personal data	=	Solely required	Condition	Data collection inspection
Compliance with applicable Ontario privacy requirements	=	Required	Condition	Security testing

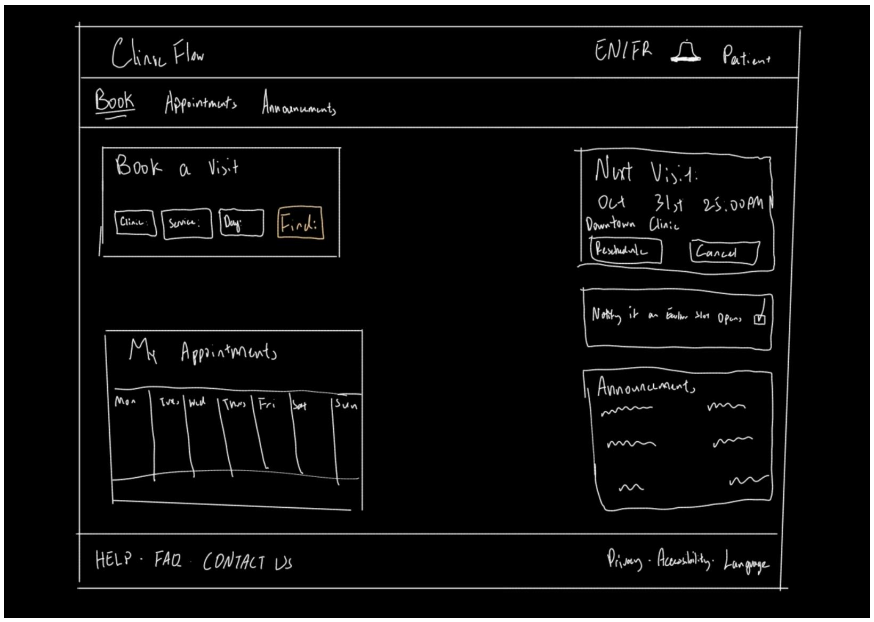
3 Global Concepts

3.1 Cameron Kibble Concept

The database and data management subsystem is responsible for securely storing, organizing and retrieving all core operational data required by ClinincFlow. This includes patient information, appointment schedules, staff availability, clinic metadata, and audit logs.

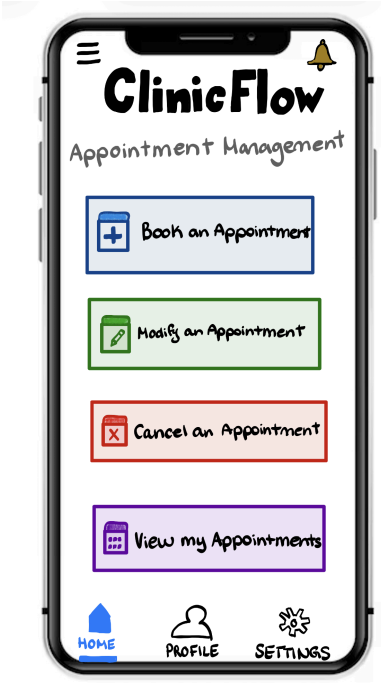


3.2 Alex Mou Concept



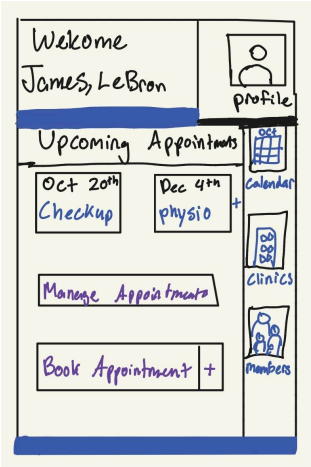
My concept is ClinicFlow's patient-facing global interface for computers and tablets. It reduces the amount of text on screen so patients can quickly find what they need (booking, their appointments and clinic announcements) without switching between separate programs or phoning the clinic. An English and French quick selection, plus contact and support links, appear on every page.

3.3 Juliana Sangregorio (Appointment Management)

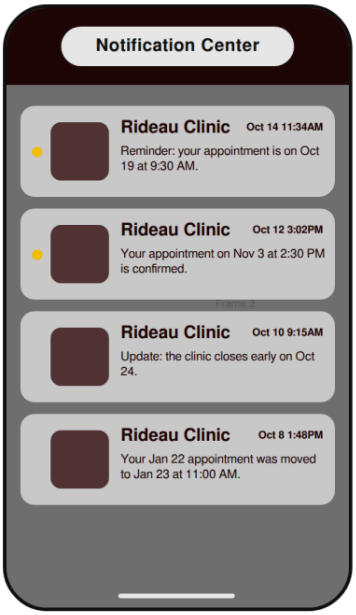


This concept uses a simple home screen with options to book, view, modify, or cancel appointments. Each option opens a separate screen for the selected task. For booking, patients select a clinic, choose an available date and time, review the details, and confirm. This approach prioritizes a straightforward booking process and accurate appointment details. Sketch to add: home screen and calendar-based booking screens.

3.4 Shyaan Qureshi Concept



3.5 Christopher Ly Concept



My concept is ClinicFlow's notification system, built into the patient home screen. When a booking, cancellation, reschedule, reminder, or clinic announcement happens, ClinicFlow would send a push alert to the patient's phone and can save the same message in a notification Center, where a bell icon with an unread badge opens. Tapping a message would open the related appointment, where the patient can interact with the appointment with whatever nessasary. Messages should be generic and have no health details to ensure privacy, and they are logged and delivered within one minute.

3.6 Analysis, Evaluation and Selection

The global concepts were evaluated based on how well they meet ClinicFlow's target specifications for usability, security, accessibility, reliability, and affordability.

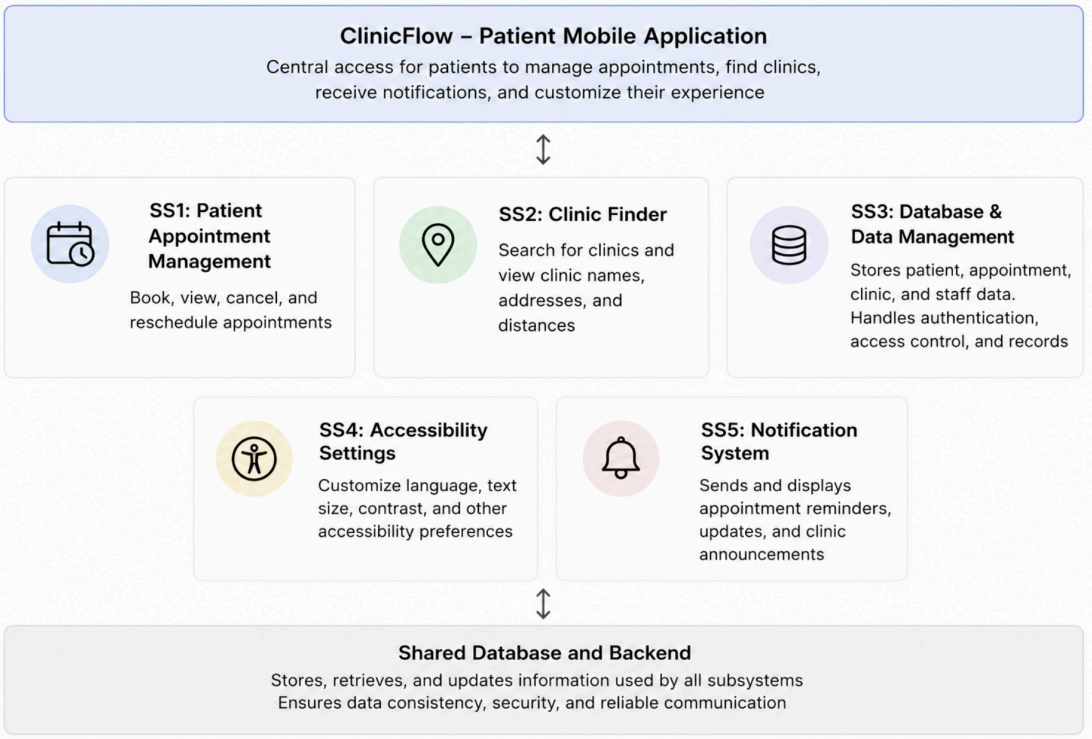
The five global concepts were compared based on how well they meet the needs of ClinicFlow. Cameron's database concept would help store and organize information, but it would need proper

security measures. Alex's patient interface would make the app easier to use, but it would need to stay simple for different users. Juliana's appointment management concept would help patients book, change, and cancel appointments, but it would need to be easy to navigate.

The selected global concept is a simple mobile application that allows patients to manage their appointments and receive notifications. It will also include accessibility settings and a secure database. This concept was selected because it covers the main needs of both patients and clinic staff while keeping the app easy to use.

4 Subsystems

ClinicFlow is divided into five subsystems with separate responsibilities. They communicate through a shared backend and database. Patient Appointment Management handles booking actions; Clinic Finder provides clinic information; Database and Data Management stores required records and controls access; Accessibility Settings stores and applies user preferences; and Notification System delivers reminders and updates. The notification subsystem sends messages based on appointment or clinic events but does not create or change appointments itself.



4.1 Subsystem 1: Patient Appointment Management

This subsystem allows patients to book, view, cancel, and reschedule appointments through the ClinicFlow app. When booking an appointment, patients select a clinic, choose a date and time, and confirm their booking. They can also view their upcoming appointments and make changes when needed. The goal is to make managing appointments simple for patients.

4.2 Subsystem 2: Clinic Finder

This subsystem helps patients find a clinic where they can book an appointment. Patients can search for clinics and view their names, addresses, and distances. After choosing a clinic, they can continue the appointment booking process. This makes it easier for patients to find a clinic that works for them.

4.3 Subsystem 3: Database and Data Management

This subsystem stores and manages the information used by ClinicFlow. This includes patient information, appointment details, clinic locations, staff availability, and system records. It also manages user login information and controls access to protect personal data. When an appointment is booked, cancelled, or rescheduled, the database is updated so that the patient and clinic staff see the correct information. This subsystem supports the other three by storing the information they need to function properly.

4.4 Subsystem 4: Accessibility Settings

This subsystem allows patients to customize the ClinicFlow app based on their needs and preferences. Patients can change the language between English and French, adjust the text size, enable high contrast mode, and use text-to-speech features. Once the settings are saved, the app automatically applies the changes. The goal is to make ClinicFlow easier to use and more accessible for patients with different needs.

4.5 Subsystem 5: Notification System

This subsystem allows patients to receive and view notifications about their appointments and clinic updates. Patients receive notifications when an appointment is booked, cancelled, or rescheduled, as well as reminders before upcoming appointments. Clinic staff can also send announcements to keep patients informed about important updates. Notifications appear in a dedicated section of the app, with the most recent notifications displayed first. The system aims

to deliver notifications within one minute of an appointment change. This helps patients stay informed and reduces the need to contact the clinic for updates.

5 Subsystem Concepts

5.1 Cameron Kibble Concepts - Database and Data Management Subsystem

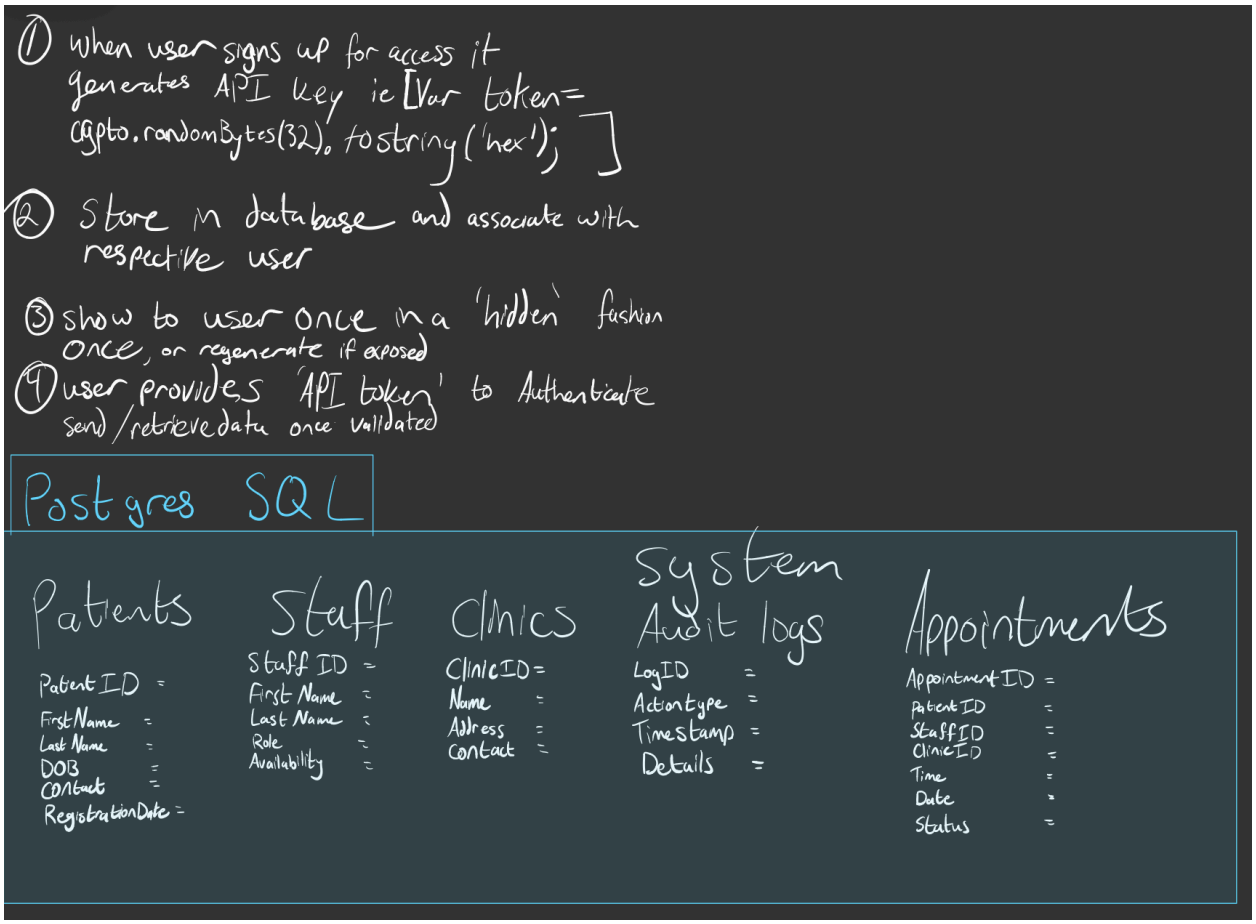
This subsystem concept implements the core backend logic, secure data storage, and authentication workflows required for ClinicFlow. It is structured into three components:

1. Secure client server communication and authentication layer
    a. Transport security: all data transfers from patient mobile app communicates via HTTPS with transport layer security (TLS) to prevent interception
    b. Authentication and authorization: using an API authentication layer with an auth token issuer. User credentials are protected using hashed/encrypted passwords, secure cookies (SameSite=strict), and support for OAuth2 with OpenID Connect or similar third party login integration.
    c. API token workflow: when a user signs up, a secure cryptographic token is generated.

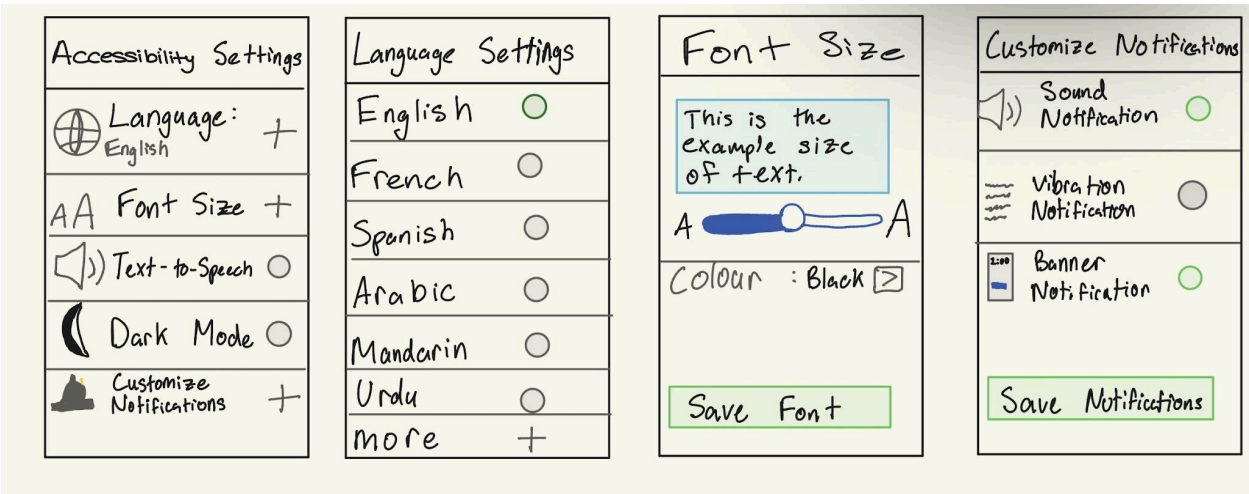
        i.e. a 32 byte random hex string [crypto.randomBytes(32).toString('hex')]. The token is then stored in the database, displayed to the user once in a hidden fashion

with options to regenerate if exposed. This API token is then required for all data transmission and retrieval requests.

2. ~~PostgreSQL~~ firebase/superbase database: the backend relies on a structured postgreSQL database containing 5 core tables to manage operations
    a. Patients: stores unique patient identifiers and profile details (PatientID, First Name, Last Name, DOB, Contact, RegistrationDate)
    b. Staff: Tracks clinic personnel details and scheduling constraints (StaffID, First Name, Last Name, Role, Availability)
    c. Clinics: Maintains facility metadata (ClinicID, Name, Address, Contact)
    d. Appointments: Links patient, staff, and clinics to manage bookings (AppointmentID, PatientID, StaffID, ClinicID, Time, Date, Status)
    e. Audit logs: Fulfills security and tracking requirements by recording system actions (LogID, ActionType, Timestamp, Details)
3. App server management:
    a. An application server processes core logic, routing requests between the database and the mobile interface to handle appointment creations, modifications, notifications, and administrative management seamlessly.



5.2 Shyaan Qureshi Concepts - Accessibility Settings

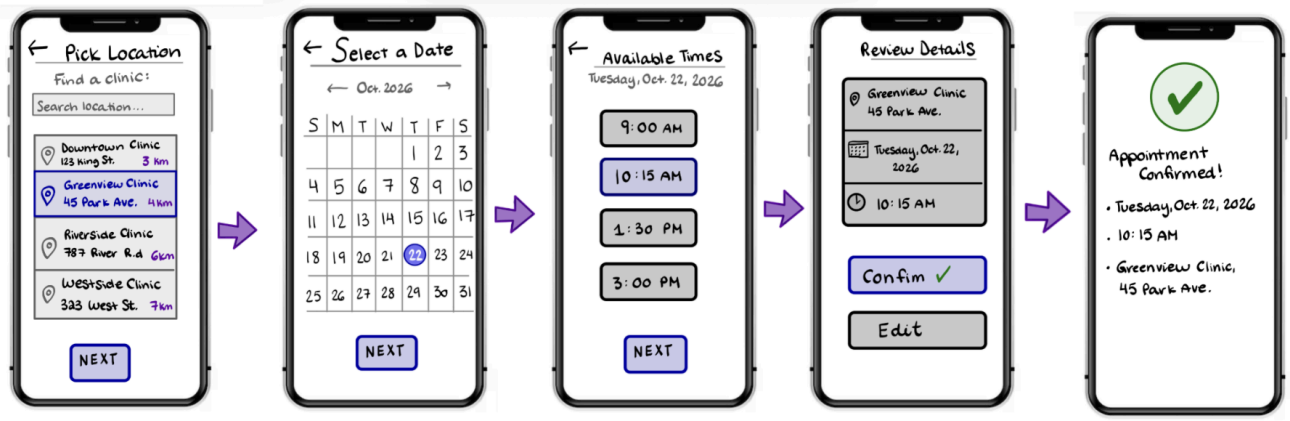


The accessibility settings subsystem allows patients to change how ClinicFlow looks and works based on their needs. Users can adjust the text size, switch between English and French, turn on high contrast mode, and use text-to-speech. These features make the app easier to use, especially for patients who may have difficulty reading smaller text or navigating certain screens.

As shown in the sketch, patients can open the accessibility settings page and choose which features they want to use. For example, they can increase the font size to make appointment details easier to read or turn on high contrast mode to make information stand out. Once they save their preferences, the app automatically applies the changes. This allows patients to customize their experience without making the app complicated.

This subsystem helps make ClinicFlow more accessible to a wider range of patients. It also supports the project's goal of creating a simple and user-friendly application that works well on mobile devices. By giving patients more control over how they use the app, ClinicFlow can make tasks like booking appointments and viewing information easier and more convenient.

5.3 Juliana Sangregorio Concepts - Patient Appointment Management & Clinic Finder



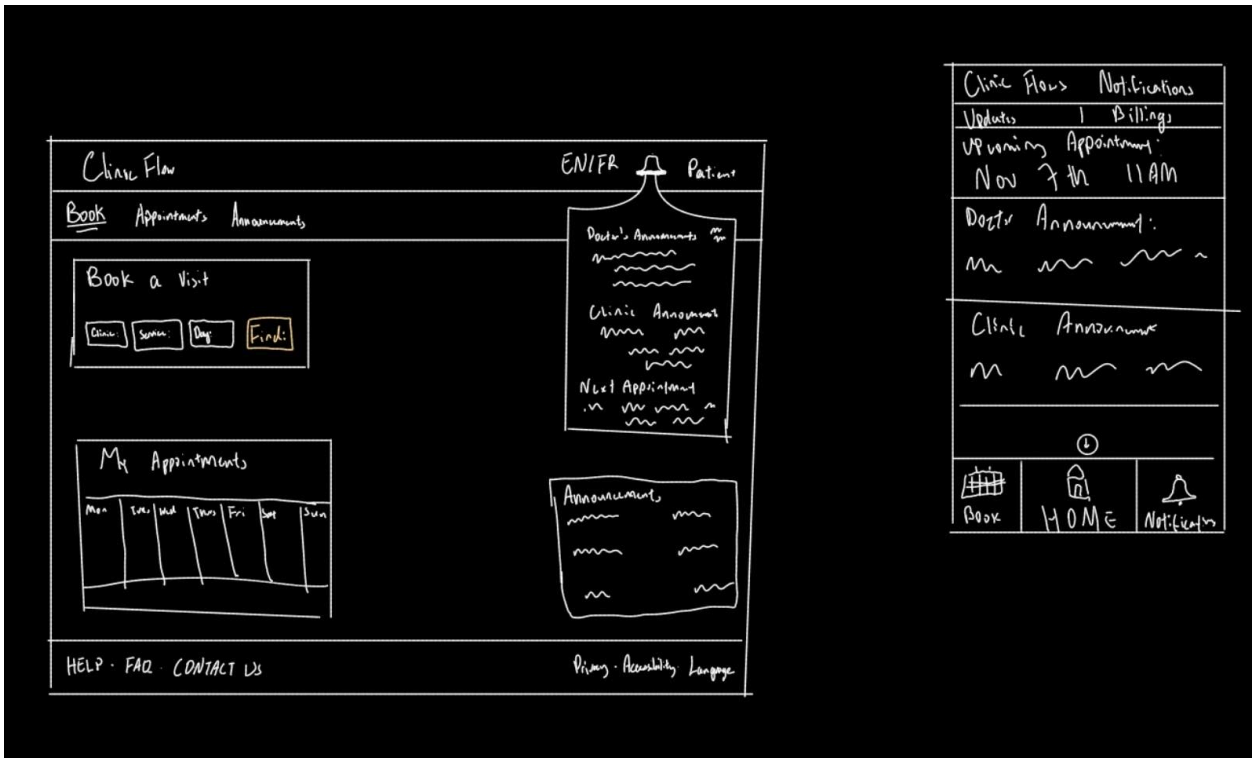
This subsystem allows patients to book appointments through ClinicFlow. It guides patients through the booking process, from selecting a clinic to confirming their appointment. The goal is to make appointment booking simple and easy to navigate while reducing the need for patients to contact the clinic directly.

Calendar-based booking:

The subsystem uses a calendar to display available appointment dates. Patients first select a clinic from a list of nearby locations, which includes the clinic's name, address, and distance. They then select a date from the calendar and choose from the available appointment times. Before confirming, patients can review the clinic location, appointment date, and time to make sure the information is correct. Once confirmed, a confirmation screen displays the appointment details.

The calendar-based design was selected because it allows patients to easily view dates and choose an appointment time that fits their schedule. The review screen also gives patients a chance to check their information before confirming their booking. Overall, this subsystem is designed to make the appointment booking process straightforward and reduce confusion for patients.

5.4 Alex Mou Concepts - Notifications subsystem

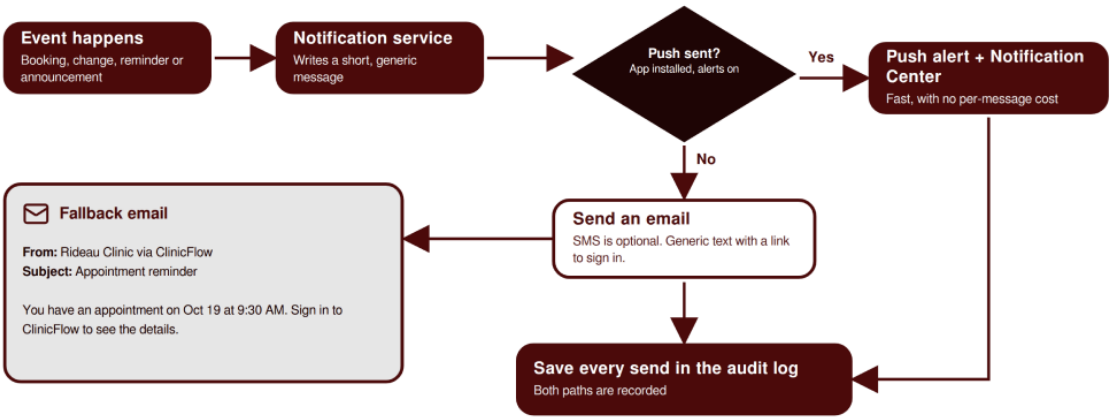


This subsystem supports patients by providing an area for them to receive and view important notices/notifications. The main goal of this subsystem is to provide a location that is easy to access and easily viewable in order for patients to constantly be kept up to date regarding personal and clinic operations.

On the left is a computer/tablet view of the notifications bar, it is accessed through the bell icon which pulls up a notification menu. In this menu, it will provide the most recent information near the top, and as we move down, less recent information can be found.

On the right is a phone view of the notifications menu, it is accessed slightly differently as a dedicated button located in the bottom navigation bar. This button will combine both important announcements and notifications along with any billings that may be occurring soon creating a hotzone of important information.

5.5 Christopher Ly Concepts



The notification subsystem decides when a notification is sent, how it reaches the patient and what it sends.This system has an email fallback, where if the push notification is not sent it would send an email and make sure to save every information sent to keep a log of all data sent.

5.6 Analysis, Evaluation and Selection

The concepts for each subsystem were compared based on their advantages, disadvantages, and how well they meet the target specifications.

For Patient Appointment Management, the calendar booking concept was selected because it lets patients view available times and book appointments. It is easy to understand, but the calendar must be organized clearly to avoid confusion.

For the Clinic Finder, the searchable clinic list was selected because it allows patients to find a clinic using basic search options. It is simple to use, but it needs accurate and up-to-date clinic information.

For the Database and Data Management subsystem, Cameron's database concept was selected because it keeps information organized and supports secure data storage. However, it needs proper security measures to protect patient information.

For Accessibility Settings, Shyaan's settings page was selected because it allows users to adjust options such as text size and language. This makes the app easier to use, but too many settings could make the page harder to navigate.

For the Notification System, Alex's notification concept was selected because it allows patients to receive reminders and updates about their appointments. However, notifications must be useful and not sent too often.

6 Conclusions

Deliverable C developed ClinicFlow's target specifications, global concept options, and five subsystem designs. The proposed application allows patients to find clinics, book and manage appointments, receive notifications, and customize accessibility settings. A shared backend supports the subsystems by maintaining consistent information and controlling access. The concepts were compared using usability, accuracy, security, accessibility, reliability, and cost as the main criteria. The selected direction combines a simple patient interface, calendar-based appointment management, clinic search, secure data management, accessibility controls, and a dedicated notification area. The remaining work is to confirm the concepts with the team and client, add the required freehand sketches, update the table of contents, and test the selected designs against the target specifications.

7 Bibliography

[1] M. Popovska, "EHR software cost in 2026: A complete pricing breakdown," Pabau, Sep. 25, 2026. [Online]. Available: https://pabau.com/blog/ehr-software-cost/. [Accessed: Oct. 4, 2026].

[2] V. Tran, "AI billing software pushed hospital costs up $942 million with no rise in care," Yahoo News, Sep. 29, 2026. [Online]. Available: https://www.yahoo.com/news/us/articles/ai-billing-software-pushed-hospital-111100239.html. [Accessed: Oct. 4, 2026].

Zoom notes:

Key Outcomes

The team decided to focus on the **patient perspective** for their design 1 and identified several **global systems** for the app, including a login page, homepage, and security features 2. They are currently refining the list of global concepts to meet rubric requirements before a client meeting 3.

Decisions Made

- **User Perspective**: The project will proceed from the **patient/client perspective** as it is more intuitive for the team 1.
- **Global Systems**: The following core features were identified as global concepts 2:
    - **Login Page**: Including user roles 2.
    - **Homepage**: Featuring appointment management (book, cancel, view, and past appointments) and availability 24.
    - **Security**: Including user authentication and calendar integration 25.

Requirements

- **Deliverable C**: Requires picking one global concept from the problem statement, providing freehand sketches, descriptions, and dividing the concept into at least **three subsystems** with boundaries, advantages, and disadvantages 36.
- **Client Meeting**: A 15-20 minute meeting with the client is pending, requiring the team to organize their findings into a slideshow 3.

Open Questions

- **Global System Definition**: There is uncertainty regarding the exact definition of a "global system" versus a "subsystem" 7.
- **Additional Concepts**: The team needs one or two more global systems to complete their list 2.

Action Items

- **Alex**: Email the TA to get clarification on the definition of global systems 7.
- **Alex**: "Lock in" to work on the project after their midterm this Saturday 3.
- **Team**: Create freehand sketches of the UI for the identified global concepts 6.

Deadlines

- **Peer Evaluation (2.1)**: Due Sunday, October 18th 8.
- **Assignment 2 (2.2)**: Due November 1st 8.

Database storage and transfer(Cameron)

Security

Appointment management(Julianna)

Notification System(Christopher)

Global UI(Alex Mou)

Power point link

https://uottawa-my.sharepoint.com/personal/jsang042_uottawa_ca/Documents/Presentation%201.pptx?d=w5ccd35fba5764b88813ad3e021b8c88d&csf=1&web=1&e=hEI9Jh

Project Deliverable Report Instructions

This document is a template for project deliverable submissions. Your group will edit a similar document for every deliverable in the course as you develop more aspects of the design process. Some elements will be repeated across deliverables. In most cases, you can copy and paste from a previous deliverable, making sure to update the text as necessary. See each deliverable's instructions for further information.

Template conventions:

- Remove or modify all red text, it is only there to guide you
- Remove this page (instructions)
- Replace all instances of <xxx> with the appropriate information for your group, for example you could replace <GROUP NUMBER> by 'B1.3'

GNG1103

Project Deliverable D

<GROUP NAME AND GROUP NUMBER>

<TEAM PHOTO>

Submitted by:

<TEAM MEMBER 1, STUDENT NUMBER>

Alex Mou, 300519307

<TEAM MEMBER 3, STUDENT NUMBER>

<TEAM MEMBER 4, STUDENT NUMBER>

<TEAM MEMBER 5, STUDENT NUMBER>

<DATE>

University of Ottawa

Table of Contents

Table of Contents....ii

1 Introduction.... 1

2 Detailed Design – Global Concept.... 2

3 Detailed Design – Subsystems.... 3

4 Bill of Materials (BOM) and Test Plan....4

5 Task Planning....5

6 Conclusions.... 6

7 Bibliography.... 7

1 Introduction

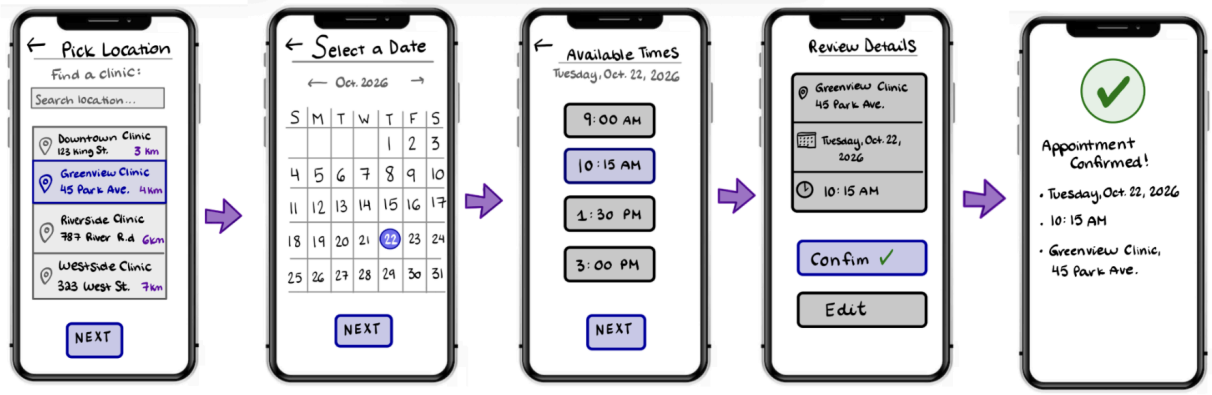
Provide the problem statement and enough context for a non-technical reader to understand your project. State any assumptions that you have made for your work. Give an overview of the structure of your document and summarize the scope of activities. Update from the previous deliverable, as required.

Canadian clinics and patients are currently relying on a fragmented administrative system in order to keep track of all their operations. Existing technologies are too bulky and complex, leading to many inefficiencies and bottlenecks within their operations. This creates a frustrating and tedious process for the patients. With costs rising, smaller local clinics suffer from high overhead costs [1], while complex billing software can drive up operational expenditure without improving patient care [2]. This makes a proper connection between patients and clinics much more difficult. This document is organized into six different sections with a reference section at the end.

Our group's main assumptions surround the stakeholders affected by our solution as we assumed that the main stakeholders will be clinics, patients and doctors.

2 Detailed Design – Global Concept

Provide a detailed drawing (understandable by an external reviewer) of the global concept chosen in the previous deliverable. Subsystems of the concept are delineated and identified. Descriptive text is provided to help clarify complex sections of the concept.



3 Detailed Design – Subsystems

Provide detailed drawings (understandable by an external reviewer) of each subsystem of the global concept. All parts and components are included in the drawings, including fasteners, adhesives, wires, power supplies, libraries, frontend/backend components, or APIs, depending on your project.

Github potentially? What coding language do we want to use?

1. Database - firebase or superbase? - Cameron

Thank you random poster from 4 years ago…

“A document store stores only keys and values. A relational DB is used to map relationships between data(primary key - foreign key).

So if i can use my hypothetical chat application as an example, it might be better to use a relational DB. You would have a table for users and groups for example. Then you have a

participation table so that you can map who belongs to what group chat. The participation table will have a ID, userID and groupID field. The userID will be the foreign key mapped to the User table's primary key and the same for the groupID.

But if you just want to store user names and passwords for example, then a document store might be more appropriate. The key is the username and the password is the value. No relationships between data is mapped. Most document stores allow you to have "buckets" which can be seen as tables in relation DB terms. Just a grouping of key-value pairs. Thats the basic gist of it.

Sometimes you can use a mix of both. For example the primary data has relationships and needs to be stored as a relational DB, but you use a document store to cache often used data.

And if you are willing to use denormalised data and to write string processing code, you can maybe use a document store for relational needs. You use the key as the "primary key" and the value is just a concatenated string of data(or a json string) that contains the rest of the "fields".

All in all, document stores are much faster and if your data doesn't have relationships(or if it isn't neccessary to map it out), then use a document store."

    a. Proper logging
        i. input/output and how that data is secured
    b. Getting information accurately
        i. speed

2. Device adaptive UI - Shyaan, Alex

    a. Update by sunday

3. Clinic finder - juliana
    a. Finding patient location
4. Calendar - Christopher
    a. Availability?

4 Bill of Materials (BOM) and Test Plan

Ensure you include all the information required in the BOM template for both the detailed design concept and the low fidelity focused prototypes and testing. Ensure you include in the Test Plan template all the required information about what you plan to test and how you will use your low fidelity focused prototypes for these tests. Provide a discussion on how you developed the information in these templates here. See the deliverable instructions for more details. Don't forget to include your filled-in templates in this deliverable's submission.

https://trello.com/b/DuA6wYqB/gng1103-project

- Claude subscription?

Since we need to provide a link to our trello board about our tasks we can create the tasks in the trello board and have them relate to the tasks we need to write for the test plans on the excel sheet

5 Task Planning

Provide a link to your Trello board or TA approved alternative here. This should include a list of tasks, duration estimates for each task, and a person identified as responsible for each task. Also include a screenshot of your most up-to-date task list here, along with a discussion of how your project tasks are progressing, problems you have encountered, or any changes in the direction of your project your team has decided on.

- List at least 3 tasks for self on trello give a due date (reasonable) for your given subtask
    - Alex and Shyaan please collaborate on UI

6 Conclusions

Provide reasonable and appropriate concluding remarks about important insights and current status of the project, as well as outstanding work remaining.

7 Bibliography

Insert your list of references here.

[1] M. Popovska, “EHR software cost in 2026: A complete pricing breakdown,” Pabau, Sep. 25, 2026. [Online]. Available: https://pabau.com/blog/ehr-software-cost/. [Accessed: Oct. 4, 2026].

[2] V. Tran, “AI billing software pushed hospital costs up $942 million with no rise in care,” Yahoo News, Sep. 29, 2026. [Online]. Available: https://www.yahoo.com/news/us/articles/ai-billing-software-pushed-hospital-111100239.html. [Accessed: Oct. 4, 2026].

Excel testplan link GNG1103_TestPlan_Template.xlsx

Excel BOM GNG1103_BOM template.xlsx

ITP review

Development goals should be as specific, actionable, and as quantifiable as possible. After each member presents their development goal, the team should offer strategies to support that member toward goal accomplishment.

Alot of development for the team can be more proactivity, reviewing requirements ahead of time.

Cameron:

- Strength: Leadership & initiative, generally taking charge of deliverable meetings, ensuring the team is on track with requirements, and completing deliverables on time.
- Goal of Development: Hold a ‘meeting/checkin’ early in the week for what tasks are required, building a checklist with assigned time estimates, getting 100% signoff on the scope for the week instead of using Thursday's meeting to do so. The meeting on Thursday should be used as a status update on tasks.
- Others: focusing on supporting team members with task delegation, maintaining clear project pacing, and attempting to foster a collaborative environment.

Juliana:

- Strength: Juliana is easy to get along with, and she brings a positive attitude to the team. She never complains, and she is always willing to help anyone out. She completes her work on time.
- Development: Juliana doesn’t share as much as the others during group discussions, especially on topics she knows less about. However, when she shares her ideas, she

brings valuable insight to the team. Her goal is to share a few suggestions during each meeting.

- Others:

Christopher

-Strengths:

Christopher has high reliability, integrity and ability specifically in reference in promoting harmony within the team, displaying high levels of empathetic concern and managing disagreements. His team highlighted his ability to contribute towards the team. Being able to deliver high quality work, staying organized and helping synthesize project ideas.

-Development:

Christopher needs to develop his ability to communicate and share information and ideas, being able to express himself more and effectively communicate to be able to stay focused on key priorities in meetings and projects. His goal is to be able to better express his thoughts and collaborate more within the team.

-Others:

Christopher has a self rating averaging one point less than his peer rating, showing how he views himself

Alex

-

-

-

Shyaan

Scope

Metrics for program:

Design specs:based on interpreted needs

Functional req: design criteria that affects function of the solution

Non func

Constraint set of guiding considerations that must be taken into account

Metric: measurable attributes ie weight size speed, resistance etc.

Metrics from Benchmarking

Testing

- Choose metrics choose what to test for prototype
- Benefits are to avoid design flaws and ease of use issues (QA)
- Can save time and money

What are our functional and non functional reqs?

How will we test these reqs

What constraints do we have

Testing user clicks: Why: to ensure it accomplishes the problem statement - reducing the headache in getting appts. What: user clicks to complete task How:have parents test the software when: 5 minutes

DELIVERABLE D MUST TEST THIS in PROTOTYPE:

- **Booking an appointment aspect, double booking protection, 100% accurate information (speed in updating the database is important too)**
- **No invalid data and no double booking - constraints**
- **Database - firebase/superbase**